

A Survey of Multimodal Hallucination Evaluation and Detection

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Abstract

Multi-modal Large Language Models (MLLMs) have emerged as a powerful paradigm for integrating visual and textual information, supporting a wide range of multi-modal tasks. However, these models often suffer from hallucination, producing content that appears plausible but contradicts the input content or established world knowledge. This survey offers an in-depth review of hallucination evaluation benchmarks and detection methods across Image-to-Text (I2T) and Text-to-image (T2I) generation tasks. Specifically, we first propose a taxonomy of hallucination based on faithfulness and factuality, incorporating the common types of hallucinations observed in practice. Then we provide an overview of existing hallucination evaluation benchmarks for both T2I and I2T tasks, highlighting their construction process, evaluation objectives, and employed metrics. Furthermore, we summarize recent advances in hallucination detection methods, which aims to identify hallucinated content at the instance level and serve as a practical complement of benchmark-based evaluation. Finally, we highlight key limitations in current benchmarks and detection methods, and outline potential directions for future research.

Keywords: Multi-modal large language model, Vision-language model, Diffusion model, Hallucination evaluation, Hallucination detection

1 Introduction

Over the past few years, Multi-Modal Large Language Models (MLLMs) have demonstrated remarkable advancements in bridging visual and textual data, supporting a wide range of multi-modal understanding and generation

tasks. Image-to-Text (I2T) models such as GPT-4o ([Open AI, 2024](#)), Gemini ([Team et al., 2023](#)), and Qwen-VL ([Wang et al., 2024](#)) excel at tasks like visual question answering and image captioning, achieving robust image recognition and reasoning capabilities without relying on external tools. Conversely, Text-to-Image (T2I) models like Stable Diffusion ([Rombach et al., 2022](#)) and

DALL-E (Ramesh et al., 2021) have made strides in generating high-quality images that align with user-specified content or artistic styles through textual prompts. Despite recent progress, both I2T and T2I models continue to face several significant challenges, such as robustness to distribution shifts (Zhang et al., 2024; Guo et al., 2024; Aafaq et al., 2022), susceptibility to attacks (Zhang et al., 2025; Fan et al., 2024), and hallucination (Huang et al., 2023; Bai et al., 2024; Lan et al., 2024; Liu et al., 2024). In particular, hallucination arises when models generate outputs that seem plausible but deviate from the given input or factual knowledge.

The hallucination problem remains a significant challenge to the development of large models. Early studies primarily focus on hallucination within I2T models, emphasizing inconsistencies between generated textual outputs and their corresponding visual inputs (Rohrbach et al., 2018; Li et al., 2023). More recent works have extended this focus to T2I models (Chen et al., 2024; Hu et al., 2023), where hallucinations manifest as misalignments between generated visual outputs and textual prompts. This survey focuses on the hallucinations across both I2T and T2I paradigms of MLLMs, categorizing them into faithfulness hallucinations and factuality hallucinations. Faithfulness hallucination involves inconsistencies between model outputs and user inputs or prior output, such as misidentified objects or images that do not correspond to textual prompts. In contrast, factuality hallucination refers to contradictions between model outputs and established world knowledge, exemplified by incorrect landmark recognition or violations of medical plausibility in X-ray imaging tasks.

As shown in Fig. 1, hallucination evaluation and benchmark construction have received increasing attention, resulting in numerous benchmarks for assessing hallucinations in both I2T models (Li et al., 2023; Wu et al., 2024; Fu et al., 2023; Seth et al., 2024; Zhou et al., 2023) and T2I models (Hu et al., 2023; Feng et al., 2023; Gokhale et al., 2022; Huang et al., 2024) from various perspectives. This paper systematically reviews existing I2T and T2I benchmarks targeting faithfulness and factuality hallucinations. Specifically, we provide a detailed comparison across benchmarks in terms of data sources, evaluation tasks, image-text pair construction, and hallucination

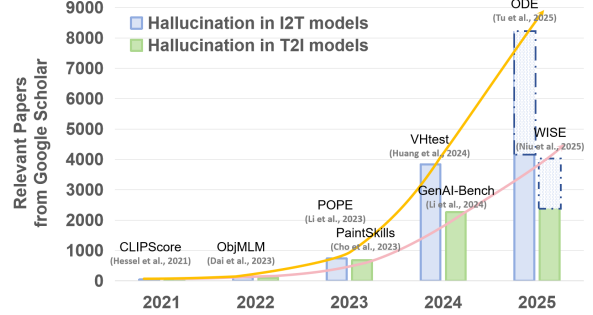


Fig. 1 Trends in the number of relevant papers on I2T and T2I hallucination evaluation from Google Scholar, highlighting the rapid growth in recent years. Dashed lines indicate approximate predictions.

types, highlighting the emerging trends towards automatic construction and fine-grained evaluation. Furthermore, we give a unified perspective on hallucination evaluation across T2I and I2T benchmarks, summarizing their commonalities as well as differences.

The flexible response formats of MLLMs present additional challenges for hallucination evaluation, particularly in free-form tasks such as image captioning and image generation. Therefore, hallucination detection methods play a crucial role in the construction of evaluation benchmarks by reducing reliance on costly human annotations and enabling scalable evaluation. Conversely, evaluation benchmarks provide annotated data and standardized metrics essential for the development and assessment of hallucination detection methods. To provide a more holistic perspective on the complementary relationship between evaluation and detection, we further provide a comprehensive summary of existing hallucination detection methods and discuss the feasibility of hallucination detection in I2T and T2I models from a unified perspective.

In a nutshell, this paper aims to advance hallucination evaluation in MLLMs through a comprehensive survey of benchmarks and detection methods for both I2T and T2I tasks. In contrast to existing surveys (Huang et al., 2023; Bai et al., 2024; Lan et al., 2024; Liu et al., 2024) that primarily focus on hallucinations arising in I2T tasks, we broaden the scope to include hallucinations in T2I tasks and provide a unified overview of both faithfulness and factuality hallucinations. Furthermore, we offer a systematic summary of existing hallucination evaluation and

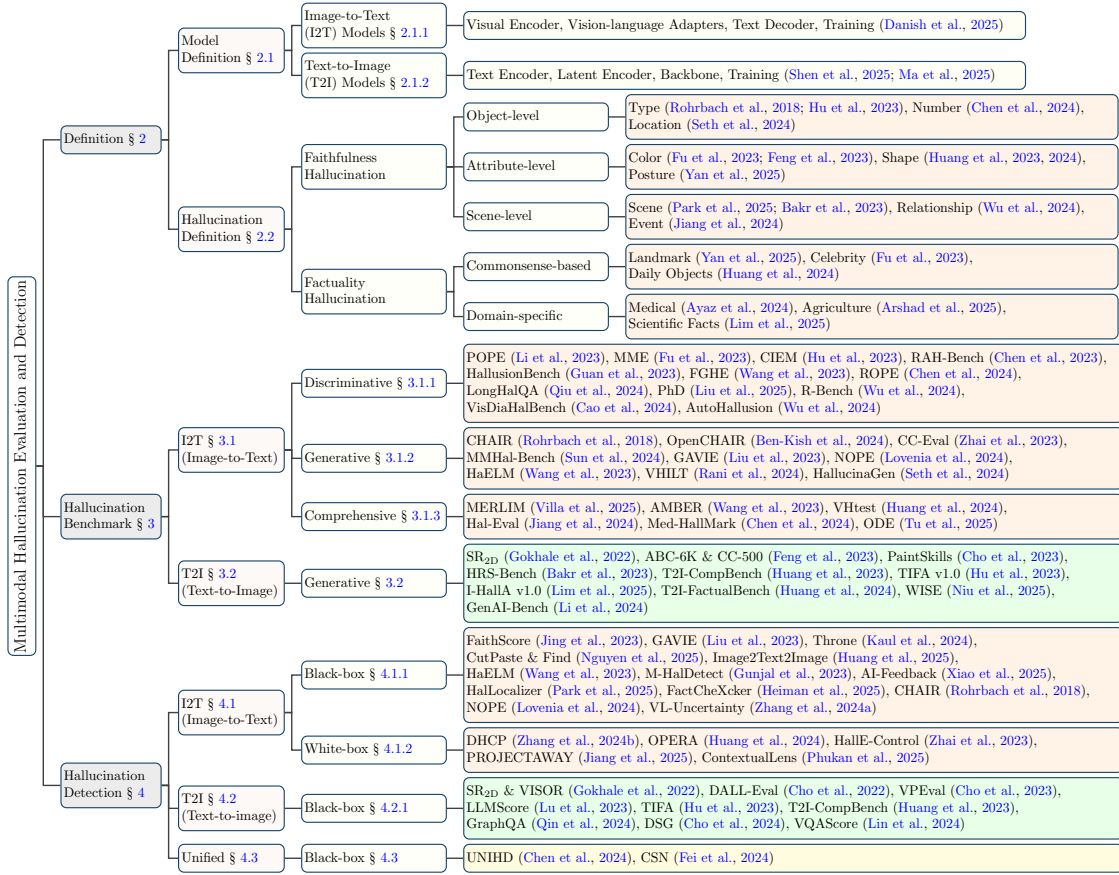


Fig. 2 Overview of the main structure and taxonomy presented in this survey.

detection methods and introduce a clear categorization framework. For example, we categorize detection methods into black-box and white-box approaches with different granularity. We also highlight recent efforts towards unified hallucination evaluation and detection across both I2T and T2I models, contributing to a more comprehensive understanding of hallucination phenomena.

Organization of this survey. To facilitate a clearer understanding of existing research, we organize this survey as follows. As shown in Fig. 2, we begin formally defining the models and the concept of hallucinations within this context in Sec. 2. Next, we analyze and summarize existing hallucination benchmarks for I2T and T2I models in Sec. 3, focusing on aspects such as hallucination types, construction methodologies, and evaluation metrics, while highlighting recent trends in benchmark development. Building on this foundation, we categorize existing MLLMs hallucination detection methods and investigate the feasibility

of unified detection approaches in Sec. 4. Finally, we discuss current limitations and challenges in hallucination evaluation, and suggest promising directions for future research.

2 Definition

In this chapter, we provide the conceptual basis by defining multimodal models and hallucination. Since hallucinations arise from complex interactions between model architecture, training dynamics, and modal alignment, we first provide precise definitions of both the I2T and T2I models and outline the components and mechanisms that may guide their behavior in Sec. 2.1. Sec. 2.2 then formalizes the definition of hallucination in MLLMs, categorizing them into faithfulness and factuality hallucinations.

2.1 Definition of Models

This section introduces the two primary categories of multimodal generative systems examined in this survey: I2T models and T2I models. Sec. 2.1.1 and Sec. 2.1.2 describe I2T and T2I models respectively, focusing on their architectures, multimodal interaction mechanisms and training. These definitions support the later discussion on hallucination causes.

2.1.1 Image-to-Text (I2T) Models

Image-to-text (I2T) models have transitioned from specialized, task-specific architectures to unified, general-purpose systems capable of complex reasoning, detailed captioning, and visual question answering (VQA). Representative models such as LLaVA (Liu et al., 2023), Qwen-VL (Wang et al., 2024), and GPT-4V (OpenAI, 2023) typically comprise a visual encoder, an alignment adapter, and an LLM decoder. Architectural constraints in these components, such as resolution limits, are distinct causal factors for hallucination. Consequently, errors arise from both inherited LLM tendencies, like over-reliance on language priors (Huang et al., 2024), and unique mismatches between visual representations and the decoder’s semantic space.

Visual Encoder. The evolution of visual encoders in I2T models reflects a shift from static, low-resolution global alignment toward dynamic, high-fidelity feature extraction intended to reduce sensory hallucination. Early contrastive architectures such as CLIP (Radford et al., 2021) treat the image as a “bag of features,” identifying objects without enforcing spatial structure. Their fixed low-resolution inputs also act as low-pass filters that remove fine details and trigger object- or attribute-level hallucinations. Recent works such as LLaVA-NeXT (Liu et al., 2024a) and Qwen2-VL (Wang et al., 2024) adopt dynamic-resolution strategies (e.g., AnyRes grid decomposition), which preserve high-frequency details by processing images at native scales. However, these solutions introduce new trade-offs: grid decomposition may break object continuity and create duplicate predictions, while variable-length sequences can overwhelm the decoder with long token streams that suffer from information dilution (Liu et al., 2024b) or visual attention sinks (Kang et al., 2025).

Vision-language Adapter. Adapter design balances information preservation and token efficiency, directly influencing hallucination behaviors. MLP-based adapters such as LLaVA (Liu et al., 2023) preserve visual detail but may induce faithfulness errors when modality gaps cause the model to rely on language priors (Liang et al., 2022). Q-former compression in BLIP-2 (Li et al., 2023) creates bottlenecks that can remove important cues and lead to non-existence hallucinations (Yan et al., 2024). Deep fusion approaches such as Flamingo (Alayrac et al., 2022) attempts closer interleaving of modalities, yet gating mechanisms may still allow the LLM’s internal knowledge to dominate, resulting in omission of visually grounded information (He et al., 2025).

Text Decoder. The text decoder often utilizes powerful decoder-only LLMs (e.g., Vicuna, LLaMA, Qwen) as the text generation backbone, a design choice that inextricably links the model’s linguistic capabilities to the root causes of hallucination (Liu et al., 2024). This reliance creates a conflict where strong pre-trained language priors (e.g., “fork” following “plate”) often override weaker visual signals, leading to object existence hallucinations (Li et al., 2023). This issue is exacerbated by “sycophancy,” a learned behavior from instruction tuning where models prioritize user intent over visual factuality, agreeing with misleading premises (e.g., confirming the presence of a non-existent object) to remain “helpful” (Pi et al., 2025). Furthermore, “parametric knowledge interference” occurs when the model’s encyclopedic pre-training supersedes specific visual context, causing it to hallucinate biography-style details or facts that are true in the world but false for the specific image at hand (Zhu et al., 2024).

Training. Training aligns visual and textual modalities, utilizing strategies ranging from single-stage adapter integration (e.g., LLaMA-Adapter (Zhang et al., 2023)) to two-stage pipelines combining feature alignment with instruction tuning (Liu et al., 2024; Zhu et al., 2024). Hallucination is often “trained into” the model through biased data or misalignment of objectives. On the data format, an exclusive reliance on positive image-text pairs fosters a stubborn “Yes-bias”, necessitating special designs in instruction tuning where models are explicitly taught to reject non-existence objects (Liu et al.,

2023). Alignment is shifting from reinforcement learning from human feedback (RLHF [Christiano et al. \(2017\)](#)) to direct preference optimization (DPO [Rafailov et al. \(2023\)](#)) and variants like HA-DPO ([Zhao et al., 2023](#)). By utilizing preference pairs that contrast accurate versus hallucinatory outputs, HA-DPO directly penalizes ungrounded tokens to enforce visual factuality rather than mere conversational fluency.

2.1.2 Text-to-Image (T2I) Models

Text-to-Image (T2I) models have evolved into high-capacity architectures capable of producing photorealistic and semantically aligned images by mapping natural language to complex visual manifolds. Modern systems, such as Stable Diffusion 3 ([Esser et al., 2024](#)) and DALL-E 3 ([Betker et al., 2023](#)), leverage massive datasets and advanced alignment techniques to achieve high semantic fidelity. Architecturally, the field is shifting from U-Net backbones toward Diffusion Transformers (DiT) ([Peebles and Xie, 2023](#)), comprising four core components: a text encoder for semantic extraction, a VAE for latent compression, a diffusion-based backbone for iterative denoising, and an alignment pipeline.

Text Encoder. Text encoders convert prompts into structured embeddings capturing semantic and syntactic information. Early models relied on contrastive encoders like CLIP ([Radford et al., 2021](#)), whose global pooling often overlooks compositional constraints ([Koishigarina et al., 2025](#)). LLM-based encoders such as T5 ([Chung et al., 2024](#)), LLaMA ([Touvron et al., 2023](#)), and Baichuan ([Yang et al., 2023](#)) provide richer syntactic awareness and better handle complex prompts, including negation, counting, and long-range dependencies ([Jiao et al., 2025](#)). Dual-stream attention mechanisms, as in MMDiT ([Stability AI, 2024](#)), enable cross-modal interaction while maintaining distinct representations, reducing faithfulness-related hallucinations.

Latent Encoder. In latent diffusion models, a Variational Autoencoder (VAE) compresses high-dimensional images into a lower-dimensional latent space ([Rombach et al., 2022](#)). The VAE outputs a latent distribution, sampled for reconstruction, with KL divergence regularizing it toward a standard Gaussian. Compression is lossy: fine textures and subtle details are discarded ([Relic et al.,](#)

2024), forcing the decoder to hallucinate missing information. Variants include standard VAEs and quantized VAEs (VQVAE ([Esser et al., 2021](#)), VQGAN ([Van Den Oord et al., 2017](#))), sometimes augmented with adversarial losses.

Backbone. The backbone predicts noise residuals or velocity fields that drive generation. While early U-Net architectures ([Ho et al., 2020](#)) suffer from structural hallucinations due to limited long-range interactions ([Lu et al., 2025](#)), the transition to Diffusion Transformers (DiT) ([Peebles and Xie, 2023](#)) mitigates this by employing global self-attention to maintain full-image receptive fields. This architecture scales effectively, as demonstrated by Flux.1 ([Black Forest Labs, 2024](#)) and Stable Diffusion 3 ([Esser et al., 2024](#)). Furthermore, state-of-the-art models increasingly adopt Rectified Flow ([Liu et al., 2022](#)) to enforce straight generation trajectories, significantly improving stability and eliminating artifactual hallucinations during few-step synthesis.

Training. Recent T2I training increasingly utilizes Rectified Flow Transformers ([Esser et al., 2024; Peebles and Xie, 2023](#)) rather than standard diffusion U-Nets, optimizing a generative backbone to straighten noise-to-data trajectories via flow matching ([Liu et al., 2022; Lipman et al., 2023](#)). The paradigm remains multi-phase: pre-training on massive image-text pairs for coarse modality alignment, followed by high-resolution finetuning. However, human alignment now favors DPO over complex RLHF pipelines, directly optimizing likelihoods against preference pairs without explicit reward modeling. For adaptation, parameter-efficient techniques like LoRA ([Hu et al., 2022](#)) and ControlNets ([Zhang et al., 2023](#)) persist but are now adapted for transformer architectures. Throughout these phases, stability is ensured via velocity-based constraints and classifier-free guidance augmentation, enabling models to generate high-fidelity images that adhere strictly to complex semantic prompts.

2.2 Definition of Hallucination

Hallucination is a general psychological term ([Bentall, 1990](#)) referring to a percept-like experience that occurs without an external stimulus, possesses the vividness of genuine perception, and is not under voluntary control ([Slade and Bentall, 1988](#)). In the context of artificial

intelligence, hallucination typically refers to the generation of unfaithful or counterfactual content. This phenomenon has garnered significant attention across multiple fields, including natural language generation (Ji et al., 2023), computer vision (Rohrbach et al., 2018; Yu et al., 2018), and multi-modal modeling (Bai et al., 2024).

In the context of computer vision, the term “hallucination” originated in image super-resolution (Baker and Kanade, 2000; Liu et al., 2007; Wang et al., 2014; Huang et al., 2019) task and has since been applied to other image generation tasks, such as inpainting (Yu et al., 2018; Quan et al., 2024) and novel sample generation (Hariharan and Girshick, 2017; Yang and Wang, 2025). In practice, hallucination is leveraged to generate realistic images or to enhance generalization capabilities. However, unlike its beneficial role in generative tasks, hallucination in vision perception tasks is often undesirable. It typically refers to instances where the model produces false or misleading outputs that do not correspond to the input data. For example, in object detection (Rohrbach et al., 2018) or image caption (Kayhan et al., 2021; Piasco et al., 2021), a model may hallucinate objects that are not present in the scene, leading to reduced reliability and potential risks in safety-critical applications such as autonomous driving (You et al., 2024) or medical imaging (Ayaz et al., 2024).

With the rapid advancement of image generation techniques, the notion of hallucination has shifted to the alignment between generated content and the input prompt (Aithal et al., 2024; Kim et al., 2024; Wang et al., 2024). Moreover, hallucination in I2T, *i.e.*, visual perception (Saleh and Tabatabaei, 2025), and T2I, *i.e.*, visual generation (Agnese et al., 2020), tasks have become increasingly interrelated: generated images are used to evaluate the faithfulness of I2T models and vice versa. One interesting work attempts to detect hallucinations across both I2T and T2I tasks (Chen et al., 2024) within a unified framework. To provide a more consistent viewpoint between T2I and I2T tasks, we adopt a unified definition of hallucination applicable to both I2T and T2I settings: *any inconsistency between the generated content and either the input conditions or established world knowledge*. This definition is consistent with the psychological concept of hallucination as a perception without an external

stimulus that contradicts reality and also provides a comprehensive viewpoint of recent works.

2.2.1 Hallucinations in I2T Tasks

Hallucination in Vision-Language Models (VLMs) refers to the generated text response being inconsistent with the input visual content or established world knowledge. Building on the taxonomy of hallucination in the context of LLM (Huang et al., 2025), we extend this framework to VLMs and categorize hallucinations into two types: faithfulness hallucination and factuality hallucination. The former captures inconsistencies between the generated contents and the visual inputs, while the latter refers to discrepancies between the generated contents and established world knowledge (Fu et al., 2023; Seth et al., 2024; Guan et al., 2023). These kinds of hallucinations may arise from factors like noisy training data, strong language priors, cross-modal misalignment, and other related issues (Huang et al., 2025). To better assess the reliability of recent VLMs, numerous studies have proposed hallucination benchmarks that evaluate different aspects of this phenomenon, including object hallucination (Li et al., 2023), relationship hallucination (Wu et al., 2024), attribute hallucination (Fu et al., 2023) and so on. Fig. 3 and Fig. 4 present examples of different kinds of hallucinations in I2T tasks.

Faithfulness Hallucinations. Recent works on faithfulness hallucinations primarily focus on object hallucination (Bai et al., 2024; Rohrbach et al., 2018; Li et al., 2023; Ben-Kish et al., 2024; Zhai et al., 2023; Lovenia et al., 2024; Dai et al., 2023) and further categorize into category (Chen et al., 2024; Wang et al., 2023; Liu et al., 2025; Wang et al., 2023), attribute (Fu et al., 2023; Hu et al., 2023; Chen et al., 2023; Qiu et al., 2024; Wang et al., 2023), and relationship (Wu et al., 2024; Hu et al., 2023; Chen et al., 2023; Wang et al., 2023; Qiu et al., 2024; Villa et al., 2025; Wang et al., 2023) hallucination. These types form a fine-to-fine-grained taxonomy. Category hallucination refers to errors in object classification. Attribute hallucination refers to the errors in attribute recognition of specific objects. Relationship hallucination requires recognizing object category and attributes, as well as objects’ interrelations. However, hallucinations such as scene

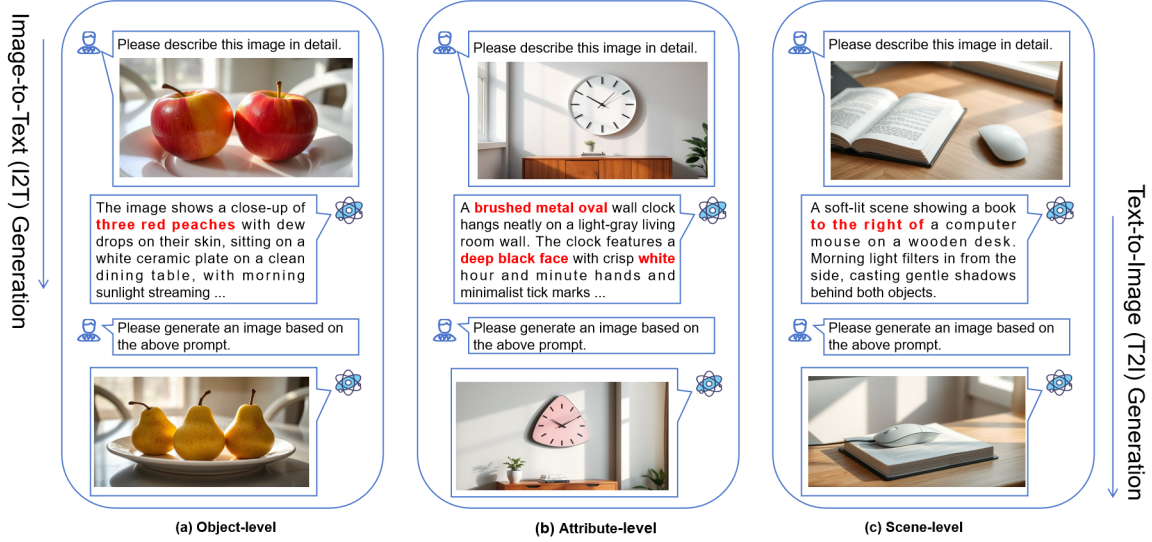


Fig. 3 Examples of object-level (a), attribute-level (b), and scene-level (c) hallucinations (from left to right) in image-to-text (top) and text-to-image (bottom) tasks. Hallucinated responses are highlighted in red. All the images are generated by Stable Diffusion 1.5 (Rombach et al., 2022).

hallucination cannot be captured by this taxonomy. Inspired by the hierarchical structure of computer vision, ranging from image-level to pixel-level based on supervision strength (Nesteruk et al., 2024), we propose a taxonomy of faithfulness hallucination based on the granularity of visual content: object-level, attribute-level and scene-level. While faithfulness hallucinations in LLMs manifesting as inconsistencies in instruction following, contextual grounding, or logical coherence (Huang et al., 2025), such granularity-based taxonomy reflects new perceptual challenges brought by multimodal.

- *Object-level* hallucinations involve the basic information of objects within the image, such as object type (Rohrbach et al., 2018), object number (Chen et al., 2024), object localization (Seth et al., 2024) and so on. For example, in Fig. 3(a), the generated description (peaches) is inconsistent with the image content (apples).
- *Attribute-level* hallucinations refer to the condition that the basic information of objects identified by VLMs is accurate, but the fine-grained intra-object content is wrong, which can be categorized into color (Fu et al., 2023), shape (Huang et al., 2024) and so on. In Fig. 3(b), the description of the clock is inconsistent with the image, particularly in terms of its material, shape, and color.

- *Scene-level* hallucinations refer to the inconsistencies between global visual inputs and textual outputs. VLMs correctly identify the basic information of objects, but hallucinate in inter-object information such as object relationships (Wu et al., 2024; Chen et al., 2023; Wang et al., 2023), object events (Jiang et al., 2024), scene (Park et al., 2025) and so on. In Fig. 3(c), the spatial relation is described as “to the right of”, which contradicts the actual positioning in the image (“to the left of”).

Factuality Hallucinations. With the advancement of in-context learning and visual understanding capabilities of VLMs, they have been increasingly applied in practical scenarios including medical image analysis (Ayaz et al., 2024), autonomous driving (You et al., 2024) and smart agriculture (Arshad et al., 2025). However, VLMs may produce hallucinations due to the insufficient grounding in established world knowledge (Seth et al., 2024; Chen et al., 2024). Existing benchmarks primarily focus on commonsense knowledge (Fu et al., 2023; Guan et al., 2023), while some recent works concerning specific domain knowledge (Seth et al., 2024; Chen et al., 2024) have emerged. Based on the source of established world knowledge, we categorized factuality hallucinations into commonsense-based and domain-specific hallucinations.

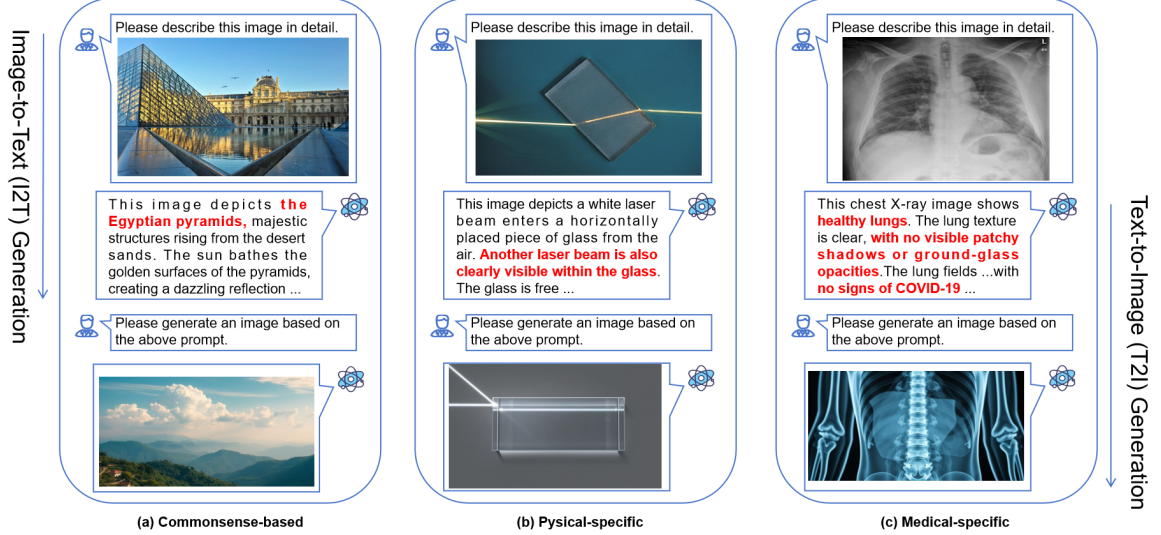


Fig. 4 Examples of Commonsense-based (a), Physical-specific (b), and Medical-specific (c) hallucinations (from left to right) in image-to-text (top) and text-to-image (bottom) tasks. Hallucinated responses are highlighted in red. All the images are generated by Stable Diffusion 1.5 (Rombach et al., 2022).

- *Commonsense-based* hallucinations refer to VLMs violating the general knowledge which is universally accepted by humans but usually implicitly stated, such as the appearance of landmarks (Yan et al., 2025) or features of celebrities (Fu et al., 2023). For example, in Fig. 4(a), VLM identifies the landmark in the image as the Egyptian pyramids, while it is actually the Louvre Museum in France.
- *Domain-specific* hallucinations refer to inconsistencies between the outputs of VLMs and factual information when processing images from specific domains. In Fig. 4(b), the VLM fails to recognize the refraction of the laser beam within the glass and instead hallucinates an additional laser beam inside the glass. In Fig. 4(c), the VLM identifies the lungs affected by COVID-19 in the X-ray image as healthy lungs.

2.2.2 Hallucinations in T2I Tasks

Hallucinations in T2I tasks refer to the generated image content being inconsistent with the input text prompt or established world knowledge. By summarizing existing T2I hallucination benchmarks (Chen et al., 2024; Feng et al., 2023; Gokhale et al., 2022; Bakr et al., 2023; Cho et al., 2023; Fei et al., 2024; Saharia et al., 2022), we categorize hallucinations in T2I tasks into faithfulness

hallucination and factuality hallucination, similar to the VLMs. Faithfulness hallucination refers to discrepancies between generated image content and input text prompt, while factuality hallucination refers to discrepancies between established world knowledge and input text prompt. Numerous studies have proposed benchmarks to evaluate faithfulness hallucination in T2I models (Feng et al., 2023; Gokhale et al., 2022; Bakr et al., 2023; Cho et al., 2023; Saharia et al., 2022), and some recent works (Huang et al., 2024; Meng et al., 2024) have also recognized the issue of factuality hallucination and introduced factuality benchmarks. Fig. 3 and Fig. 4 present examples of different kinds of T2I hallucinations.

Faithfulness Hallucinations. We conduct the same taxonomy of faithfulness hallucinations in T2I tasks like I2T tasks: object-level, attribute-level and scene-level. Recent works primarily focus on object-level (Cho et al., 2023) and attribute-level (Hu et al., 2023; Feng et al., 2023) hallucination, while some studies have explored scene-level hallucination, such as relationship hallucination (Huang et al., 2023, 2024).

- *Object-level* hallucinations refer to cases where T2I models generate incorrect basic object information during image synthesis, including object type (Hu et al., 2023), object number (Cho et al., 2023) and so on. In Fig. 3(a), the

fruits in the generated image are pears rather than peaches in the text prompt.

- *Attribute-level* hallucinations refer to cases where T2I models properly generate the object-level content, but the attributes of objects are wrong, such as color (Feng et al., 2023), shape (Huang et al., 2023), and so on, which have been explicitly specified in the text prompt. In Fig. 3(b), the generated image depicts a triangular clock, which does not match the “oval” shape specified in the text prompt.
- *Scene-level* hallucinations refer to T2I models incorrectly generating large-scale image content or inter-object information, including scene (Bakr et al., 2023), object relationship (Huang et al., 2023, 2024) and so on. In Fig. 3(c), the spatial relationship between the computer mouse and the book in the generated image is “above” rather than “to the right of”.

Factuality Hallucinations. T2I models can generate realistic images according to text prompts, but the image content often contradicts established world knowledge. Recent works primarily focus on commonsense knowledge (Huang et al., 2024; Niu et al., 2025), such as the appearance of well-known landmarks or specific types of animals. Hallucinations involving domain-specific knowledge have received relatively little attention. Similar to VLMs, factuality hallucinations in T2I models can also be categorized into commonsense-based and domain-specific hallucinations.

- *Commonsense-based* hallucinations refer to cases where T2I models generate image content that conflicts with basic commonsense knowledge, such as the features of a dog or soccer ball (Huang et al., 2024). Fig. 4(a), the generated image depicts a mountain landscape, whereas the prompt is expected to evoke an image of the pyramids.
- *Domain-specific* hallucinations arise when the image content produced by T2I models deviates from established domain knowledge, such as physical laws (Niu et al., 2025), scientific facts (Lim et al., 2025) and so on. In Fig. 4(b), the image generated by the T2I model not only fails to depict the refraction of the laser beam within the glass, but also shows an incorrect reflection angle on the glass surface. In Fig. 4(c), the image presents an abdominal X-ray instead

of the expected chest X-ray, indicating an inaccurate medical understanding.

2.3 Relationship between Hallucination Problems in I2T and T2I tasks

Building on the proposed definitional and classificatory framework, we can more clearly characterize the relationships and differences between hallucinations in I2T and T2I models. Under a unified definition, both faithfulness and factuality hallucinations emerge in both task settings, driven largely by shared factors such as cross-modal misalignment and insufficient or biased training data. Because faithfulness hallucinations stem from image-text alignment issues, I2T and T2I models exhibit similar patterns across object-, attribute-, and scene-level granularities.

Regarding factuality hallucinations, research has examined commonsense errors in both I2T and T2I tasks, while domain-specific hallucinations show clearer differences across model types. In I2T, studies increasingly focus on specialized domains such as medicine (Seth et al., 2024; Chen et al., 2024; Hartsock and Rasool, 2024) and smart agriculture (Arshad et al., 2025), supported by domain-specific benchmarks and metrics. Many I2T models pretrained on large-scale medical datasets (Ye and Tang, 2025) are evaluated using coarse-grained generative metrics (*e.g.*, BERTScore (Zhang et al., 2020)) or fine-grained discriminative metrics (*e.g.*, accuracy, precision) (Seth et al., 2024; Hartsock and Rasool, 2024). In contrast, research on domain-specific hallucinations in T2I remains limited. Medical T2I models, typically GAN- or diffusion-based (Uzunova et al., 2019; Pinaya et al., 2022; Polamreddy et al., 2024), are evaluated with general image quality metrics such as FID (Heusel et al., 2017) and CLIPScore (Hessel et al., 2021). These metrics primarily measure overall image fidelity or semantic similarity but provide limited insight into fine-grained factual correctness (Koishigarina et al., 2025; Jayasumana et al., 2024), especially in safety-critical domains. Overall, domain-specific evaluation is expanding for I2T, and developing fine-grained metrics for T2I represents a key direction for future research.

Table 1 Summary of representative evaluation metrics for hallucination in MLLMs.

Metric	Formula	Usage	Derivatives
Accuracy	$\frac{TP + TN}{TP + FP + TN + FN}$	Accuracy measures the overall correctness of answers to YNQs or MCQs	Accuracy+ (Fu et al., 2023)
Precision	$\frac{TP}{TP + FP}$	Precision measures the proportion of predicted positives that are actually correct.	—
Recall	$\frac{TP}{TP + FN}$	Recall measures the proportion of actual positives that are correctly identified.	—
F1 Score	$\frac{2 \times \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$	F1 Score measures the harmonic mean of precision and recall to balance false positives and false negatives.	$F1_{PhD} = 2 \cdot \frac{\text{Yes Recall} \times \text{No Recall}}{\text{Yes Recall} + \text{No Recall}}$ (Liu et al., 2025) Composite = MAE/F1 Score (Heiman et al., 2025)
CHAIR _s (Rohrbach et al., 2018)	$\frac{ \{\text{hall. sents.}\} }{ \{\text{all sentences}\} }$	CHAIR _s measures the proportion of sentences containing any hallucinated objects.	Attack Success Rate (ASR) (Wu et al., 2024) OpenCHAIR (Ben-Kish et al., 2024)
CHAIR _i (Rohrbach et al., 2018)	$\frac{ \{\text{hall. obj.}\} }{ \{\text{all obj.}\} }$	CHAIR _i measures the proportion of hallucinated objects among all mentioned objects.	AMBER = $\frac{1}{2} \times (1 - \text{CHAIR} + \text{F1})$ (Wang et al., 2023)
GPT Rate (Sun et al., 2024)	0-6 rating (6 = no hallucination)	GPT Rate uses GPT as an evaluator to assess the degree of hallucination.	GAVIE (Liu et al., 2023), Fine-tuned model rate (Wang et al., 2023) FaithScore = $\frac{\sum_{i=1}^C \sum_{j=1}^{n_i} w_j^i \cot s(e_j^i, I)}{\sum_{i=1}^C n_i}$ (Jing et al., 2023)
Visor (Gokhale et al., 2022)	$\frac{\text{Correct images}}{\text{Total images}}$	Visor measures the proportion of images with correct spatial relationships.	UniDet (Huang et al., 2023)
VQAScore (Li et al., 2024)	Alignment score	VQAScore evaluates the alignment between image and text using a VQA model.	TIFA Score (Hu et al., 2023) I-HallA Score (Lim et al., 2025) WiScore (Niu et al., 2025) LLMScore (Lu et al., 2023)

3 Hallucination Evaluation Benchmarks and Metrics

In this chapter, we provide a systematic overview of the evolution of existing benchmarks. Tab. 1 summarizes metrics for hallucination evaluation, including their formula and derivatives. Tab. 2 summarizes key details of these benchmarks, including their data sources, construction paradigms, and evaluation criteria. Hallucination benchmarks for I2T and T2I models are introduced in Sec. 3.1 and Sec. 3.2 respectively, with their interrelation explored in Sec. 3.3.

3.1 Hallucination Benchmarks for I2T Models

Effectively evaluating hallucination in VLMs remains an open challenge. Recent I2T benchmarks assess hallucination across diverse task formats, including Yes or No Questions (YNQs),

Multiple-Choice Questions (MCQs), Image Captioning (IC), and Visual Question Answering (VQA). These tasks are typically categorized by response type into two groups: discriminative (YNQs and MCQs) and generative (IC and VQA) (Lan et al., 2024; Bai et al., 2024). Following this convention, we categorize hallucination benchmarks based on whether they constrain the output space (discriminative) or allow open-ended generation (generative).

In addition, some benchmarks integrate both discriminative and generative tasks to analyze the inconsistencies in hallucination behavior across different evaluation tasks (Jiang et al., 2024; Wang et al., 2023), which has not been sufficiently addressed in previous surveys (Lan et al., 2024; Bai et al., 2024). To capture this feature, we introduce a third category, namely comprehensive benchmarks. This classification provides a clearer framework for understanding the design and focus of existing hallucination evaluations. We present

Table 2 Summary of representative evaluation benchmarks for hallucination in MLLMs, with task types classified as discriminative (Dis), generative (Gen), and comprehensive (Comp) tasks.

Task	Benchmark	Benchmark Type	Data Source	Size	Hallucination Type		Metric
					Faithfulness	Factuality	
I2T	POPE (Li et al., 2023)	Dis	MSCOCO (Lin et al., 2014)	3000	✓	✗	Acc/P/R/F1
	MME (Fu et al., 2023)	Dis	MSCOCO (Lin et al., 2014)	1277	✓	✓	Acc
	CIEM (Hu et al., 2023)	Dis	MSCOCO (Lin et al., 2014)	72941	✓	✗	Acc/P/R/F1
	RAH-Bench (Chen et al., 2023)	Dis	MSCOCO (Lin et al., 2014)	3000	✓	✗	FP
	HallusionBench (Guan et al., 2023)	Dis	Website	1129	✓	✓	Acc
	FGHE (Wang et al., 2023)	Dis	MSCOCO (Lin et al., 2014)	200	✓	✗	Acc/P/R/F1
	ROPE (Chen et al., 2024)	Dis	MSCOCO (Lin et al., 2014) & ADE20K (Zhou et al., 2017)	5000	✓	✗	Acc
	LongHalQA (Qiu et al., 2024)	Dis	Visual Genome (Krishna et al., 2016) & Object365 (Shao et al., 2019)	6485	✓	✗	Acc
	R-Bench (Wu et al., 2024)	Dis	MSCOCO (Lin et al., 2014)	11651	✓	✗	Acc/P/R/F1
	VisDiaHalBench (Cao et al., 2024)	Dis	GQA (Hudson and Manning, 2019)	25000	✓	✗	F1/EM
	AutoHallusion (Wu et al., 2024)	Dis	MSCOCO (Lin et al., 2014) & DALL-E-2	1000	✓	✗	ASR/MASR/CASR
	PhD (Liu et al., 2025)	Dis	TDIUC (Kafle and Kanan, 2017)	102564	✓	✓	F1 _{PhD}
	CHAIR (Rohrbach et al., 2018)	Gen	MSCOCO (Lin et al., 2014)	5000	✓	✗	CHAIR
	CC-Eval (Zhai et al., 2023)	Gen	Visual Genome (Krishna et al., 2016)	100	✓	✗	CHAIR/Cover
	HaELM (Wang et al., 2023)	Gen	MSCOCO (Lin et al., 2014)	5000	✓	✗	LLM Rate
	GAVIE (Liu et al., 2023)	Gen	Visual Genome (Krishna et al., 2016)	1000	✓	✗	GPT Rate
	OpenCHAIR (Ben-Kish et al., 2024)	Gen	Stable Diffusion	5000	✓	✗	OpenCHAIR
	MMHal-Bench (Sun et al., 2024)	Gen	OpenImages (Kuznetsova et al., 2020)	96	✓	✗	GPT Rate
	NOPE (Lovenia et al., 2024)	Gen	OpenImages (Kuznetsova et al., 2020)	36000	✓	✗	Acc/METEOR
	VHILT (Rani et al., 2024)	Gen	Website	2000	✓	✓	Acc
	HallucinaGen (Seth et al., 2024)	Gen	MSCOCO (Lin et al., 2014) & NIH Chest X-ray (Wang et al., 2017)	96000	✓	✓	Acc
	AMBER (Wang et al., 2023)	Comp	Website	15202	✓	✗	Acc/CHAIR
	VHtest (Huang et al., 2024)	Comp	MSCOCO (Lin et al., 2014) & DALL-E-3	1200	✓	✗	Acc
	Hal-Eval (Jiang et al., 2024)	Comp	MSCOCO (Lin et al., 2014)	10000	✓	✗	Acc/Score
	Med-HallMark (Chen et al., 2024)	Comp	Slake (Liu et al., 2021), etc.	7341	✓	✓	MediHall Score
	MERLIM (Villa et al., 2025)	Comp	MSCOCO (Lin et al., 2014)	31373	✓	✗	Acc/F1
	ODE (Tu et al., 2025)	Comp	Stable Diffusion	8786	✓	✓	AMBER/Acc
T2I	SR _{2D} (Gokhale et al., 2022)	Gen	MSCOCO (Lin et al., 2014)	25280	✓	✗	Visor Score
	DrawBench (Saharia et al., 2022)	Gen	Human, DALL-E	200	✓	✗	Human
	ABC-6K & CC-500 (Feng et al., 2023)	Gen	MSCOCO (Lin et al., 2014)	6900	✓	✗	Human/CLIP-R
	PaintSkills (Cho et al., 2023)	Gen	Template	7330	✓	✗	DETR Score
	HRS-Bench (Bakr et al., 2023)	Gen	GPT	15000	✓	✗	P/AC-T2I
	T2I-CompBench (Huang et al., 2023)	Gen	MSCOCO (Lin et al., 2014) & Template & GPT	6000	✓	✗	BLIP-VQA/UniDet
	TIFA v1.0 (Hu et al., 2023)	Gen	MSCOCO (Lin et al., 2014)	4000	✓	✗	TIFA Score
	T2I-FactualBench (Huang et al., 2024)	Gen	GPT	3000	✓	✓	VQA Score
	GenAI-Bench (Li et al., 2024)	Gen	Human	1600	✓	✗	Human/VQAScore
	I-HallA v1.0 (Lim et al., 2025)	Gen	Textbook	200	✓	✓	I-HallA Score
	WISE (Niu et al., 2025)	Gen	LLM-Constructed	1000	✓	✓	WiScore

representative benchmarks within each category in the following sections.

3.1.1 Discriminative Benchmarks

A growing body of benchmarks adopts discriminative tasks for hallucination evaluation. Compared with generative settings, VLMs tend to show more stable performance on discriminative tasks (Li et al., 2023), which also enable fine-grained diagnosis of different hallucination types via diverse YNQs and MCQs. As more hallucination categories emerge and LLMs gain stronger in-context learning abilities, discriminative benchmarks have expanded beyond object hallucinations, with increasingly automated construction pipelines. Considering these developments, we summarize discriminative benchmarks along two axes: their coverage of hallucination types and

their data generation strategies, highlighting differences in data sources, construction methods, and evaluation metrics.

POPE (Li et al., 2023) is an early YNQ-based benchmark for assessing object hallucinations in VLMs. It selects 500 MSCOCO (Lin et al., 2014) validation images containing at least three annotated objects and constructs questions of the form “Is there a {object} in the image?”. For “Yes” queries, annotated objects are used; for “No” queries, non-existent objects are sampled via three strategies: random sampling, popular sampling (common objects), and adversarial sampling (frequent co-occurrence objects). Evaluation metrics include Accuracy, Precision, and Recall. Model performance degrades from random to popular and adversarial settings, reflecting training data biases. While POPE is widely used for evaluating basic object-existence hallucinations, its

closed-form YNQs and single-object focus limit its alignment with open-ended real-world scenarios.

ROPE (Chen et al., 2024) extends hallucination evaluation in VLMs from single- to multi-object scenarios using MCQs. Models must identify correct object names from a list across multiple bounding boxes, making the task more challenging than POPE. Images are drawn from MSCOCO (Lin et al., 2014) and ADE20K (Zhou et al., 2017), focusing on those with more than five annotated objects and bounding boxes with IoU below 0.1. The benchmark is divided into four subsets: homogeneous, heterogeneous, adversarial, and in-the-wild. Results show that hallucination likelihood increases with the number of object categories, indicating that object co-occurrence bias remains a key driver of I2T hallucinations. ROPE has become a standard benchmark for multi-object hallucination analysis, with its dense-object and co-occurrence settings better reflecting complex real-world scenes.

FGHE (Wang et al., 2023) builds upon the POPE framework and extends the scope of hallucination evaluation to include not only object hallucination, but also relationship, attribute, and behavior hallucinations using YNQs. This benchmark selects 50 images from MSCOCO (Lin et al., 2014), and human annotators construct 200 binary YNQs based on these images. Evaluation is performed using standard metrics such as Accuracy, Precision, F1 score, and other relevant indicators, enabling a more comprehensive analysis of fine-grained hallucination types in VLMs.

MME (Fu et al., 2023) evaluates VLMs from perceptual and cognitive perspectives across various hallucination types. Perception tasks include coarse-grained recognition of objects, quantity, color, and location, and fine-grained tasks requiring external knowledge, such as movie posters, celebrity photos, and OCR. Cognition tasks cover commonsense reasoning, math, translation, and code. Images are drawn from public datasets like MSCOCO (Lin et al., 2014), and YNQs are manually constructed. The Accuracy+ metric based on Yes-No paired questions mitigates Yes/No bias, offering a reliable assessment. MME’s broad coverage approximates real-world multimodal demands and remains widely used.

HallusionBench (Guan et al., 2023) is a comprehensive benchmark evaluating two hallucination types based on visual necessity: Visual

Dependent (requiring image analysis for reasoning) and Visual Supplement (reliant on external knowledge). By utilizing manually edited images and a Yes/No bias test, the benchmark assesses model robustness and quantitative preferences. Results indicate that even advanced VLMs struggle (*e.g.*, GPT-4V achieving only 31.42% accuracy) revealing a significant tendency for models to prioritize internal parametric memory over actual visual context.”

CIEM (Hu et al., 2023) targets hallucinations regarding object existence, attributes, and relationships through an automated benchmark construction pipeline. By leveraging MSCOCO (Lin et al., 2014) annotations and GPT-3.5, it generated over 72,000 balanced YNQs, achieving a scale significantly surpassing manual benchmarks. While this method relies on pre-existing annotations, limiting its application to raw images, human verification confirmed a low 5% error rate, validating the efficacy of LLM-driven generation.

RAH-Bench (Chen et al., 2023) evaluates three types of hallucinations: object classification, attributes, and relationships. This work collects images from the MSCOCO validation set and utilizes GPT-4 to generate 3000 YNQs based on annotations. Unlike works (Li et al., 2023; Hu et al., 2023) evaluating all YNQs using the Accuracy metric, RAH-Bench introduces False Positive (FP) Rates to assess the model’s performance across different hallucination types. While Accuracy reflects overall correctness, False Positive Rates provide a more fine-grained diagnostic signal. Experiments show that VLMs perform worse in attribute and relationship hallucinations, suggesting targeted improvements.

R-Bench (Wu et al., 2024) addresses the scarcity of relationship hallucination evaluations by introducing a large-scale benchmark (11,651 instances) covering existence, spatial relationships, and actions. It utilizes a hierarchical structure with image-level (existence) and instance-level (specific types) to enable fine-grained analysis. Constructed using scene graph parsers on MSCOCO (Lin et al., 2014) and LLM-generated YNQs based on Nocaps (Agrawal et al., 2019), R-Bench highlights the specific vulnerability of VLMs to relational reasoning errors.

AutoHallusion (Wu et al., 2024) is an automated benchmark pipeline for evaluating object existence and spatial relationship hallucinations in

VLMs. It uses text prompts to probe the model’s language priors (*e.g.*, “We have {object1} in the image. What is the object to insert/remove under the strategy X where we wish to...”) and performs targeted image editing based on the responses. The pipeline employs three induction strategies: abnormal object insertion, paired object insertion, and correlated object removal. Hallucination detection with GPT-4V-Turbo achieves 92.6% accuracy. Experiments show induction success rates of 97.7% on synthetic datasets and 98.7% on real-world datasets. AutoHallusion demonstrates that exploiting VLMs’ language priors can amplify hallucination rates and provides a scalable method to systematically probe datasets for induced hallucinations.

PhD (Liu et al., 2025) evaluates hallucinations in VLMs across object existence, attributes, and counting. It uses images and annotations from TDIUC (Kafle and Kanan, 2017) and generates word sets for different question types. For color questions, GPT creates alternative color words excluding the correct one, and CLIP measures similarity between the image and descriptions. GPT then generates binary Y/NQ questions based on the highest similarity keyword, increasing difficulty compared to direct GPT question generation. PhD also employs hallucination induction via inaccurate prefixes and counterfactual images. To mitigate bias from models’ “Yes” or “No” preference, a new $F1_{\text{PhD}}$ score is adopted (as it shown in Tab. 1). PhD emphasizes challenging question design, induction methods and careful evaluation metrics for hallucination evaluation.

LongHalQA (Qiu et al., 2024) is a benchmark of 6,000 long texts covering hallucinations in object existence, attributes, counting, and relationships. Unlike short-context questions in previous benchmarks, it introduces object-level description, image-level description, and multi-round conversation formats, providing rich contexts and reflecting real-world interactions. GPT-4V generates MCQs for both discriminative and generative tasks: description selection and continuation selection. Results show model rankings in LongHalQA align with free-generation settings, demonstrating its effectiveness. LongHalQA highlights VLMs’ limitations in reasoning and grounding over extended text and offers a unified framework for LLM-free evaluation.

Summary of Discriminative Benchmarks. As shown in Tab. 2, discriminative benchmarks show clear trends in evaluation focus and construction methods. The focus has shifted from basic faithfulness hallucinations to factuality hallucinations involving commonsense and domain knowledge, moving from perceptual to cognitive assessment. Early benchmarks rely on manually designed questions, while recent efforts such as CIEM (Hu et al., 2023) and RAH-Bench (Chen et al., 2023) use LLMs for automated generation. Frameworks like AutoHallusion (Wu et al., 2024) and PhD (Liu et al., 2025) enable automated pipelines and diverse hallucination induction methods. LongHalQA (Qiu et al., 2024) unifies generative and discriminative tasks via MCQs, a creative benchmark construction method.

Currently, research on factuality hallucinations in discriminative benchmarks remains limited, with a predominant focus mainly on commonsense knowledge and insufficient coverage of specialized domains. To better align with real-world application scenarios, future discriminative benchmark construction can incorporate richer contextual formats such as multi-turn dialogues and long-text. Additionally, exploring more hallucination induction methods similar to PhD (Liu et al., 2025) could also enhance the complexity of discriminative benchmarks.

3.1.2 Generative Benchmarks

Generative tasks produce longer, open-ended responses, challenging traditional accuracy-based metrics and requiring dedicated hallucination detection. Early work focused on object hallucinations in image captioning (Rohrbach et al., 2018), while recent benchmarks leverage large language models for contextual understanding and long-form generation (Ben-Kish et al., 2024; Zhai et al., 2023). Concerns over data leakage (Sun et al., 2024; Rani et al., 2024) have prompted the use of alternative data sources, such as website-crawled or synthetic images (Ben-Kish et al., 2024; Rani et al., 2024), to improve evaluation robustness. This section reviews generative hallucination benchmarks, highlighting diverse hallucination types, expanded task formats, and automated detection methods, with attention to question types, data sources, and evaluation strategies.

CHAIR (Caption Hallucination Assessment with Image Relevance) (Rohrbach et al., 2018) is one of the early works evaluating object hallucination in the image captioning task. It measures the proportion of hallucinated objects in VLM responses with two variants: per-instance (CHAIR_i) and per-sentence (CHAIR_s), as shown in Tab. 1. Detection uses sentence tokenization and synonym mapping, and analysis is limited to the 80 MSCOCO (Lin et al., 2014) object types.

OpenCHAIR (Ben-Kish et al., 2024) is an open-vocabulary benchmark designed to evaluate object hallucination in the image captioning task. Compared to CHAIR (Rohrbach et al., 2018), OpenCHAIR expands the evaluation scope to include a wider range of rare and diverse objects, reflecting the variability of real-world scenarios. To construct the benchmark, LLM is used to generate diverse captions, and then a diffusion model generates the corresponding images. Additionally, OpenCHAIR replaces CHAIR’s fixed synonym lists with a dynamic semantic matching method based on LLMs, enabling open-vocabulary and flexible object hallucination evaluation.

CC-Eval (Zhai et al., 2023) evaluates object hallucination in detailed image captions using GPT-4 assistance. It uses 100 randomly sampled images from Visual Genome (Krishna et al., 2016), and requires VLMs to generate captions. GPT-4 extracts mentioned objects and compares them with the ground-truth annotations from Visual Genome. The benchmark adopts CHAIR to measure hallucination rate and introduces coverage to assess how well real objects are captured. These metrics jointly offer a comprehensive assessment of object-level accuracy.

NOPE (Lovenia et al., 2024) evaluates object hallucinations in cases where the answer should be “None” or “Nothing” (NegP data). To address the scarcity of such cases, it constructs 36,000 examples using two LLM-based methods: generate-from-scratch and list-then-rewrite. Generate-from-scratch prompts the LLM directly, while list-then-rewrite uses multi-turn dialogue. Human evaluation shows 50% accuracy for the former and 92% for the latter. NOPE employs Accuracy and METEOR, revealing that VLMs are prone to hallucinations on non-existent objects, reflecting a loss of negative data during training.

VHILT (Rani et al., 2024) collects 2,000 images from the New York Times Twitter account, with

VQA and caption tasks manually designed via Amazon Mechanical Turk. It defines eight hallucination types, including contextual guessing, identity incongruity, and geographic erratum. Evaluation relies on human annotations, showing models frequently hallucinate on contextual guessing and identity incongruity, highlighting limitations in visual cognition and reasoning. Due to its manual annotation dependence, future work could explore using large models to automate parts of the benchmark.

MMHal-Bench (Sun et al., 2024) collects images from the validation set of OpenImages (Kuznetsova et al., 2020) to prevent data leakage. The benchmark contains 96 handcrafted VQAs, and covers 8 types of hallucinations, including object existence, attributes, counting, and environment. Evaluation leverages GPT-4 to analyze and score the degree of hallucination based on the consistency with actual image content. Results show that GPT-4’s hallucination scores align with human annotations in 94% of cases, demonstrating the effectiveness of using LLMs for evaluation.

HaELM (Wang et al., 2023) evaluates VLMs’ performance on image captioning tasks utilizing a fine-tuned LLM. To construct the training dataset for fine-tuning, GPT is prompted to create 10,000 hallucinatory and 10,000 non-hallucinatory captions based on the MSCOCO (Lin et al., 2014) training set. LLaMA is then fine-tuned with LoRA to create a hallucination detector. For evaluation, 5,000 samples from MSCOCO test set are used. Results show that HaELM achieves 95% of GPT’s accuracy in hallucination detection. HaELM provides a scalable and cost-efficient solution without dependence on external APIs.

HallucinaGen (Seth et al., 2024) evaluates VLMs’ object recognition and reasoning capabilities through challenging visual-language tasks with increasing difficulty: Localization (LOC), Visual Context (VC), and Counterfactual (CF). Images are sourced from MSCOCO (Lin et al., 2014) and NIH Chest X-ray (Wang et al., 2017). Similar to NOPE (Lovenia et al., 2024), it employs implicit hallucination induction attacks, including locating absent entities and imagining non-existent objects. Evaluation combines word-matching algorithms with GPT-4o, using five prompts for robust assessment. Results show that as task difficulty increases, VLMs exhibit higher

hallucination rates. In tasks involving domain-specific knowledge, even specialized models like LLaVA-Med struggles to perform better than random guessing. This highlights the vulnerability of VLMs in high-stakes scenarios like healthcare, where accuracy and reliability are critical.

Summary of Generative Benchmarks.

Current generative benchmarks have advanced in diversifying hallucination types, expanding task formats, and introducing automation. However, several challenges remain. Many benchmarks still rely on coarse metrics such as hallucination rates and LLM scores, lacking fine-grained assessment. Data leakage risks persist even with web-crawled or validation-set data. Automation is mostly limited to question generation and depends on human verification. Developing scalable pipelines that combine reliable automation with diagnostic evaluation is a key direction.

3.1.3 Comprehensive Benchmarks

Considering the limited scope of discriminative benchmarks and the challenges of generative benchmarks, comprehensive benchmarks combine both approaches for a more holistic evaluation. We review existing benchmarks focusing on three aspects: broader hallucination types, more automated and scalable evaluation methods, and mitigation of data leakage risks. As shown in Tab. 1 and Tab. 2, each benchmark is summarized in terms of hallucination types, evaluation metrics, data sources, and core design features.

MERLIM (Villa et al., 2025) evaluates existence, relation, and counting hallucinations using a large number of both edited and original images. GPT is used for automatic question generation. To construct the benchmark, the original images are edited by removing a selected object and seamlessly inpainting the missing region to preserve visual realism, creating paired edited versions for controlled comparison. By comparing model responses between the original and edited images, hallucinations lacking visual grounding can be effectively identified. Relationship and counting hallucinations are evaluated via YNQs. Object hallucinations are evaluated via open-ended questions, while model responses are processed with spaCy library and GPT, followed by calculating metrics including Precision.

AMBER (Wang et al., 2023) evaluates hallucinations in object existence, attributes, and relations through discriminative and generative tasks. Images are manually collected to prevent data leakage. Discriminative questions reveal that models like GPT-4V and Qwen-VL favor “No” answers, while generative tasks extract target objects from captions and compute metrics such as CHAIR, though limited by extraction errors and focus on existence hallucinations. Unlike MERLIM, AMBER introduces a unified AMBER Score (as shown in Tab. 1) to assess overall performance, providing a holistic benchmark for model reliability across tasks.

VHtest (Huang et al., 2024) is a synthetic benchmark designed to evaluate eight types of visual hallucinations, including object existence, shape, color, and OCR, across 1,200 instances. To ensure image diversity and prevent data leakage, it constructs the dataset by first identifying initial hallucination instances from MSCOCO (Lin et al., 2014) using CLIP and DINO, and then generating additional images with T2I models based on annotations. Question construction utilizes a combination of templates and manual editing to create YNQs and image captioning tasks. YNQ responses are automatically evaluated, while captions require manual assessment. Experimental results show that this benchmark effectively reduces the accuracy of state-of-the-art models like GPT-4V and LLaVA-1.5.

VisDiaHalBench (Cao et al., 2024) includes 25,000 multi-turn questions and 5,000 edited images, covering hallucinations such as object existence, attributes, relations, and verification questions. Built on GQA (Hudson and Manning, 2019), it edits images and scene graphs by removing, replacing, or changing object colors. The modified scene graphs are used to prompt GPT-4 to generate five-turn dialogues, enabling in-depth evaluation of VLMs’ visual understanding and consistency. Evaluation applies keyword-filtered outputs to compute macro-average F1 and Exact Match scores. Results show GPT-4 performs significantly worse on VisDiaHalBench, allowing fine-grained analysis of reasoning and consistency errors that single-turn evaluations could miss.

Hal-Eval (Jiang et al., 2024) provides a systematic quantitative analysis of data leakage in VLMs. It constructs evaluation datasets from

MSCOCO (Lin et al., 2014) (in-domain) and various web sources (out-of-domain) and generates questions with GPT-4. Hal-Eval covers discriminative and generative tasks, addressing hallucinations in existence, attributes, relations, and event types. Experiments show most models favor “Yes” in discriminative tasks, revealing a bias toward positive responses. Models like LLaVA exhibit stronger hallucinations out-of-domain, likely due to extensive in-domain fine-tuning. Generative task evaluation uses a fine-tuned Hal-Evaluator, performing comparably to GPT-4 and effectively identifying hallucinations across datasets.

Med-HallMark (Chen et al., 2024) is a hallucination detection benchmark for the multimodal medical domain, supporting Med-VQA and Imaging Report Generation tasks. It evaluates model robustness using four question types: conventional, confidence-weakening, counterfactual, and image depiction, constructed via manual annotation, model-generated responses from LLaVA-Med and GPT, and prompt engineering. To characterize medical hallucinations, it introduces a hierarchical categorization with five levels—catastrophic, critical, attribute, prompt-induced, and minor—based on clinical impact. The MediHall Score quantifies hallucination severity at the answer level for Med-VQA and the sentence level for IRG. Med-HallMark provides structured metrics for rigorous safety assessment and supports systematic evaluation of multimodal models to enhance reliability in clinical applications.

ODE (Open-set Dynamic Evaluation) (Tu et al., 2025) is a dynamic benchmark for assessing hallucinations in VLMs by generating novel image-text pairs with diverse objects and attributes. It represents objects and their co-occurrence using a weighted graph and produces test samples across standard, long-tail, random, and fictional distributions. This design mitigates data leakage and allows iterative updates to expand evaluation scope. ODE employs generative and discriminative tasks with specific inquiry templates, and evaluates performance using accuracy, CHAIR, and AMBER. Experiments show ODE effectively reveals hallucination patterns and supports fine-tuning to enhance model reliability.

Summary of Comprehensive Benchmarks. As shown in the analysis and Tab. 2, comprehensive benchmarks remain relatively scarce.

While efforts like MERLIM (Villa et al., 2025) and Hal-Eval (Jiang et al., 2024) incorporate both task types, fully unified evaluation frameworks that consistently integrate results across different formats are still in progress. Benchmarks such as AMBER (Wang et al., 2023) introduce unified metrics, though achieving balanced coverage of hallucination types across tasks remains a challenge. Nevertheless, several benchmarks provide valuable strategies for mitigating data leakage. For instance, Hal-Eval (Jiang et al., 2024) includes both in-domain and out-of-domain datasets to assess potential data leakage. VHtest (Huang et al., 2024) and ODE (Tu et al., 2025) utilize T2I generation models to construct synthetic datasets, improving scalability and reducing the risk of data leakage. These developments underscore the need for more comprehensive benchmarks.

Summary of I2T Hallucination Benchmarks. Our review of I2T hallucination benchmarks reveals several trends. The range of hallucination types is expanding, LLMs are increasingly used in benchmark construction, and data leakage concerns are receiving more attention. However, despite these advancements, notable imbalances remain. Benchmarks focusing on faithfulness are more developed, while those targeting factuality hallucinations remain limited (fewer than 5 I2T benchmarks in Tab. 2). Research continues to emphasize image-text consistency, leaving broader factuality issues underexplored.

As shown in Tab. 2, discriminative benchmarks outnumber generative ones, as predefined answer sets simplify hallucination detection. Generative benchmarks are less common due to the difficulty of assessing open-ended outputs. Similarly, most benchmarks focus on a single task, with fewer supporting both discriminative and generative evaluations. Developing comprehensive benchmarks is an important future direction. Moreover, current benchmarks mainly focus on single-turn tasks including YNQs or image captioning, with fewer benchmarks addressing complex tasks like multi-turn dialogue (Cao et al., 2024). Future benchmarks could explore richer interaction formats to better reflect real-world applications.

Analysis of I2T Hallucination Causes. Understanding why hallucinations occur in I2T models requires dissecting the contributing factors beyond merely measuring them. We attribute I2T

hallucinations to two primary sources: (1) *Data Limitations*, which include insufficient quantity, low diversity, noise, and bias in the training data; and (2) *Model-Specific Mechanisms*, which encompass architectural constraints, attention dynamics, and cross-modal misalignment. This structure highlights how both the inputs and the model design jointly shape hallucination patterns, providing a systematic lens for analysis.

- *Lack of Data.* Insufficient quantity of training data remains a fundamental cause of hallucinations in I2T models. Udandarao et al. (2024) show that current datasets leave vast semantic spaces uncovered due to the exponential scaling required for zero-shot mastery. This sparsity is compounded by distributional imbalances: models struggle with underrepresented long-tail concept (Parashar et al., 2024) and unseen multi-object combinations (Chen et al., 2024), while a lack of negative samples impedes negation handling (Lovenia et al., 2024). These findings suggest that data sparsity limits the model’s ability to ground descriptions in visual evidence, pushing it to fall back on language priors, increasing the likelihood of hallucination.
- *Noisy Data.* Beyond scarcity, the quality of I2T training data is compromised by inherent noise in both web-scraped and synthetic datasets. Yu et al. (2024) identify a cascade of errors where LLM-generated instruction data contains systematic hallucinations, frequently fabricating objects, relations and attributes. When models are fine-tuned on this corrupted supervision, they inevitably internalize these hallucinatory patterns. These kinds of noisy data directly degrades the models’ ability to ground text in visual evidence, leading to persistent factual inconsistencies in downstream generation.
- *Bias in Data.* Biases in training data, such as frequency imbalance and spurious co-occurrence, also contribute significantly to hallucinations. Models systematically over-predict dominant classes (e.g., human), a tendency confirmed by the high false-positive rates (Li et al., 2023; Dai et al., 2023). Additionally, POPE (Li et al., 2023) finds that high-frequency objects increase hallucination rates in negative-sample tests. Rigid co-occurrence patterns also override visual evidence (Leng et al., 2024), e.g., labeling a black banana as yellow. These failures

indicate a fundamental reliance on statistical shortcuts rather than faithful visual grounding, causing models to hallucinate plausible but absent details when inputs deviate from training norms of I2T models.

- *Model Architecture.* The architecture of I2T models, including visual encoders and attention mechanisms, plays a crucial role in hallucination generation. Perceptual bottlenecks in visual backbones (e.g., coarse resolution in CLIP-ViT/16) cause aliasing, where distinct details are compressed into ambiguous embeddings (Radford et al., 2021; Tong et al., 2024; Wei et al., 2024). Attention misallocation prevents effective utilization of these features. Models frequently underutilize visual data even when available (POPEv2, Li et al. (2025)) and fail to ground generated tokens in relevant image regions (Xie et al., 2025). Autoregressive decoding process further amplifies these initial slips: as generation progresses, early grounding errors accumulate, causing the model to drift further from visual evidence in longer sequences (Zhou et al., 2024).
- *Cross-modal Misalignment.* Hallucinations frequently stem from the fundamental difficulty in synchronizing disparate modalities (Wu et al., 2025). Misalignment stems from distributional gaps in the joint space (Liu et al., 2025) that lead to erroneous object-attribute associations (Alonso et al., 2025). Furthermore, models exhibit a structural bias toward suppressing visual influence and causing the model to hallucinate based on language probabilities rather than image (Guan et al., 2023; Huang et al., 2024; Zhou et al., 2024; Zheng and Zhang, 2025). While recent works propose enhancing visual salience (Xie et al., 2024), modulating textual signals (Li et al., 2025), or cross-modal verification (Huang et al., 2024), misalignment remain a key structural source of hallucination.

3.2 Hallucination Benchmarks for T2I Models

While I2T benchmarks have advanced rapidly, hallucination evaluation in T2I models has also progressed in hallucination types, benchmark construction, and evaluation methods. Early works focused on image-text alignment, such

as color (Feng et al., 2023) and spatial relationships (Gokhale et al., 2022), categorized as faithfulness hallucinations. Recent studies (Huang et al., 2024) extend to factuality hallucinations using prompts based on world knowledge. Benchmark construction has become more automated, evolving from manual dataset sampling (Feng et al., 2023) to template-based or LLM-assisted generation (Huang et al., 2023; Bakr et al., 2023). Evaluation methods have similarly shifted from manual annotation (Saharia et al., 2022) and object detection (Gokhale et al., 2022) to automated approaches using VLMs, VQA, and captioning tasks (Huang et al., 2023; Hu et al., 2023), enabling fine-grained and scalable assessments.

The following section provides an overview of existing T2I hallucination benchmarks, highlighting hallucination types, data sources, and evaluation metrics. As shown in Tab. 2, T2I benchmarks are generally generative, assessing image generation correctness.

SR_{2D} (Gokhale et al., 2022) is a benchmark designed to evaluate spatial hallucination in T2I models. Based on 80 object categories from MSCOCO (Lin et al., 2014), the benchmark contains a total of 25,280 text prompts. The VISOR metric is used to calculate the accuracy of the generated images. It extracts spatial relationships from generated images using object detection and centroid determination algorithms. Experimental results show that T2I models exhibit poor performance in understanding and rendering spatial relationships, as indicated by low VISOR scores.

ABC-6K & CC-500 (Feng et al., 2023) are two benchmarks targeting color hallucination in T2I models. ABC-6K constructs 6400 text prompts by sampling descriptions from MSCOCO (Lin et al., 2014) and swapping color words in those that contain at least two color terms. CC-500 focuses on simple parallel text prompts, typically in the format of “a red apple and a yellow banana.” The evaluation includes human evaluation, automatic evaluation using the phrase alignment model GLIP, and system-level metrics such as FID (Heusel et al., 2017). Experimental results reveal three main challenges in generating images from combined prompts: attribute leakage, interchanged attributes, and missing objects.

DrawBench (Saharia et al., 2022) is a comprehensive benchmark including a diverse set of 200 prompts across 11 categories. These categories test

various aspects of model performance, including color, quantity, spatial relationships, scene text, and unusual object interactions, as well as complex prompts with long descriptions, rare words, and misspellings. Human evaluations are conducted by presenting images from different models and asking raters to compare them in terms of image fidelity and image-text alignment.

PaintSkills (Cho et al., 2023) assesses image-text alignment in T2I tasks using 7330 template-generated text prompts. It evaluates hallucinations such as object existence, counting, and spatial relationships. Text prompts are generated through templates, and over 50000 images are created using Unity engine to train an object detector (DETR). The evaluation employs DETR to automatically verify object presence in the generated outputs, and determines spatial relationships through algorithmic analysis. Results show that T2I models generally perform poorly in object counting and spatial relationships.

HRS-Bench (Bakr et al., 2023) is a scalable evaluation benchmark for T2I models, assessing accuracy, robustness, generalization, fairness, and bias. It covers 50 diverse scenes across domains such as fashion, animals, transportation, and objects. Unlike prior methods relying on manual prompt design, HRS-Bench constructs specific hallucination templates (*e.g.*, “Describe a scene about 3 apples and 3 dining tables”) and expands them using GPT into more complex prompts (*e.g.*, “Three apples are sitting on top of three dining tables in a cozy dining room”). Evaluation uses the AC-T2I metric, which captions the generated image and computes alignment with the original text prompt with an I2T model.

T2I-CompBench (Huang et al., 2023) is a comprehensive open-world benchmark for compositional T2I generation. It contains 6,000 text prompts across three main categories (attributes, relations, complex compositions) and six subcategories (color, shape, texture, spatial, etc.), derived from templates, existing datasets, or ChatGPT generation. Evaluation uses a multi-tool approach: BLIP-VQA assesses object attributes, UniDet evaluates spatial relationships, and a mean metric combining CLIPScore, BLIP-VQA, and UniDet scores measures complex compositions. Additionally, VLMs (*e.g.*, MiniGPT-4) assess overall image-text alignment. This approach improves the precision of hallucination detection in T2I models.

TIFA v1.0 (Hu et al., 2023) constructs a text prompt dataset for evaluation by sampling from the image descriptions of MSCOCO (Lin et al., 2014), containing 4K text inputs. It utilizes a language model to pre-generate corresponding question-answer pairs for each prompt, covering diverse elements. This process yields 25K questions spanning 4.5K element types. Moving beyond coarse-grained evaluation, TIFA employs VQA tools for fine-grained assessment across 11 hallucination categories (*e.g.*, object type, count, attributes). The resulting VQA accuracy is reported as the TIFA Score. Experiments show that TIFA Score has a higher correlation with human judgments than CLIPScore.

I-HallA v1.0 (Lim et al., 2025) constructs a dataset of real image-text pairs sourced from scientific and historical textbooks, with a focus on factuality hallucinations. The texts are enriched with external knowledge and hallucination reasoning using GPT-4’s pre-trained capabilities. An LLM then generates MCQs based on the augmented texts, while T2I models are tasked with producing corresponding images. During evaluation, a vision-language model answers MCQs associated with each image-text pair, and the I-HallA Score is calculated based on its accuracy. This MCQ-based framework offers great evaluation stability compared to caption.

T2I-FactualBench (Huang et al., 2024) evaluates factuality hallucinations in generated images. It covers 1,600 knowledge concepts across eight domains and defines three prompt difficulty levels: basic factual combinations (*e.g.*, objects sized according to real-world facts), instantiated factual combinations (*e.g.*, a table with specific foods), and instantiated knowledge concepts with interactions (*e.g.*, a dog playing fetch with a hat). Evaluation uses multi-round VQA and assigns three scores: Concept Factuality, Task Completeness, and Composition Factuality, enabling granular assessment of factuality hallucinations.

WISE (World Knowledge-Informed Semantic Evaluation) (Niu et al., 2025) evaluates factuality hallucinations in T2I tasks using complex prompts across natural sciences, spatiotemporal reasoning, and cultural knowledge. It contains 1,000 prompts spanning 25 subfields, sourced from educational materials, encyclopedias, common sense Q&A datasets, and LLM-generated data. Human

annotators refine the prompts for clarity, complexity, and unambiguous real-world answers. Evaluation uses the WiScore metric, a weighted sum of GPT-4o-assigned scores for consistency, realism, and aesthetic quality, providing a semantically rich assessment of image generation.

GenAI-Bench (Li et al., 2024) is a T2I benchmark with 1,600 manually constructed text prompts, covering common hallucination types such as attributes, spatial relations, and actions. GenAI-Bench also introduces complex real-world scenarios involving counting, comparisons, and textual negation. All prompts are refined with the assistance of GPT. Evaluation combines human scoring with VQAScore metric. VQAScore quantifies image-text consistency by computing the probability of a “Yes” response from a VQA model to the question “Does this figure show ‘{text}’?”. VQAScore exhibits higher correlation with human than CLIPScore on GenAI-Bench, validating its effectiveness for hallucination assessment.

Summary of T2I Hallucination Benchmarks. Recent evolution in T2I benchmarks highlights three key trends. First, the scope has broadened from semantic faithfulness to external factuality, evaluating adherence to both prompt constraints and world knowledge (*e.g.*, T2I-FactualBench Huang et al. (2024)). Second, the involvement of LLMs enables automated benchmark construction. In HRS-Bench (Bakr et al., 2023), text prompts are generated based on templates first and then expanded by GPT. Third, leveraging VLMs facilitates fine-grained evaluation, exemplified by TIFA v1.0 (Hu et al., 2023), which employs VQA models to detect object- and attribute-level hallucinations.

Despite recent progress, several challenges remain. Although some pretrained T2I models in specialized domains have emerged (Abaid et al., 2024), existing hallucination benchmarks still pay insufficient attention to domain-specific hallucinations. Moreover, many benchmarks still rely on coarse-grained scoring provided by VLMs (Niu et al., 2025; Li et al., 2024). Exploring more fine-grained evaluation methods represents a promising direction for future research.

Analysis of T2I Hallucination Causes. While I2T and T2I models share some common sources of hallucination, such as biases and insufficiency in training data, T2I models exhibit distinct challenges due to the generative nature

of image synthesis. These include difficulty handling negative prompts, structural inconsistencies caused by iterative generation, and pixel-level attention ambiguities that can blend or omit objects. We organize the causes into two categories, emphasizing the factors most specific to T2I generation: (1) *Data Limitations* and (2) *Model-Specific Mechanisms*.

- *Data Limitations.* Mirroring I2T limitations, T2I models are affected by data sparsity and bias. In particular, negative prompts such as “the bathroom area *without* a curtain” are underrepresented in common datasets (Bui et al., 2025), *i.e.*, MSCOCO 0.43% (Lin et al., 2014), InstructPix2Pix 0.02% (Brooks et al., 2023), CC3M 1.63% (Sharma et al., 2018), causing T2I models to ignore or incorrectly generate negated objects (Conwell et al., 2024). Beyond negation, coverage gaps in complex compositional and factual scenarios frequently lead to generative failures (Huang et al., 2023; Lim et al., 2025). Moreover, distributional biases in object co-occurrence, style, or demographics frequently override prompt instructions, resulting in stereotyped or fabricated outputs (Vice et al., 2025).
- *Model Architecture and Attention.* T2I hallucinations are closely tied to specific architectural vulnerabilities. The text encoder acts as a semantic bottleneck, and pre-trained encoders like CLIP often exhibit “bag of words” behavior, leading to incorrect comprehension of complex prompts (Koishigarina et al., 2025). The U-Net backbone with long skip connections can propagate unrefined noise from early layers to the decoder, bypassing semantic guidance and producing structural inconsistencies such as disjointed limbs or incoherent textures (Huang et al., 2023). Meanwhile, diffusion dynamics often fail to separate semantically distant concepts, leading to mode interpolation and hybrid objects (Aithal et al., 2024). Finally, cross-attention failures, such as attribute bleeding and token suppression, cause fine-grained object omissions and bleeding (Chefer et al., 2023; Phung et al., 2024). These visual manifestations distinguish T2I hallucinations structurally from the textual errors observed in I2T models.

3.3 Relationship between Hallucination Benchmarks for I2T and T2I Models

Although hallucination evaluation originated earlier and is more comprehensive in I2T tasks, T2I benchmarks centered on image-text consistency are now gaining increasing attention. Both directions share similar trends, including the broadening of hallucination categories, increasing automation in benchmark construction, the use of model-based evaluation methods, and a stronger focus on fine-grained analysis. However, existing benchmarks capture only a subset of real-world hallucination scenarios. They are invaluable for controlled studies, but they often simplify the task (*e.g.*, binary questions (Guan et al., 2023), limited answers (Zhai et al., 2023)) and restrict the domain (*e.g.*, common objects in single images (Li et al., 2023; Chen et al., 2024)). Recent benchmarks are beginning to fill these gaps by considering multi-turn dialogues (Cao et al., 2024), long-text (Qiu et al., 2024), and hallucination induction (Liu et al., 2025) in I2T and open-world generation (Huang et al., 2023) and real-world scenarios (Li et al., 2024) in T2I. These advances aim to better capture the complexity of hallucinations encountered in real-world applications. Broadening hallucination evaluation to incorporate factual content and world knowledge remains an important direction for future work.

In terms of construction and evaluation strategies, there is a growing overlap in the use of models across both tasks. For instance, I2T benchmarks such as VHtest (Huang et al., 2024) and PhD (Liu et al., 2025) employ T2I models to generate or modify images for building datasets. Conversely, T2I benchmarks like TIFA v1.0 (Hu et al., 2023) and T2I-FactualBench (Huang et al., 2024) use VLMs to evaluate hallucinations. These practices suggest two complementary developments: the use of synthetic image data in I2T benchmarks, and the application of VLM-based evaluation in T2I benchmarks. They highlight the potential for mutual enhancement between I2T and T2I research in advancing hallucination evaluation.

3.4 Hallucinations of MLLMs in Key Domains

MLLMs show great promise, yet they face significant reliability issues in safety-critical contexts. This section analyzes the occurrence and mitigation of hallucinations across high-stakes domains, such as embodied AI, medical diagnostics, remote sensing, and agriculture.

- *Embodied AI.* I2T models increasingly drive manipulation, navigation, and autonomous systems (Jiang et al., 2022; Majumdar et al., 2022; Guo et al., 2024). Frameworks like VIMA (Jiang et al., 2022) and Robotool (Xu et al., 2023) generate autoregressive plans for unseen objects and creative tool use, while MUTEX (Shah et al., 2023) enables adaptive human-robots interaction. These capabilities extend to assistive technologies, with systems like WalkVLM (Yuan et al., 2024) and EgoBlind (Xiao et al., 2025) further extend VLMs to visually impaired assistance, providing real-time navigation and egocentric understanding for the visually impaired. Despite these advancements, hallucinations constitute a critical safety bottleneck, frequently distorting scene understanding or triggering unsafe actions. This risk is particularly acute in autonomous driving, where benchmarks like NuScenes-QA (Qian et al., 2024) reveal that grounding failures often cap model accuracy below 70% (Yin et al., 2021; Jiao et al., 2023). To address this, emerging mitigation strategies target both signal processing and reasoning. At the signal level, Long et al. (2025) apply truncated SVD to filter hallucination-prone residuals, boosting traffic captioning accuracy to 87%. Procedurally, multi-turn reasoning frameworks like Socratic Planner (Shin et al., 2024) and RAT (Wang et al., 2024) employ iterative self-correction and retrieval-augmented thought to verify plans and ensure robust execution.
- *Medical Imaging.* There have been a large number of medical I2T models, such as MedFlamingo (Moor et al., 2023), RadFM (Wu et al., 2025), HuatuoGPT-Vision-7b (Chen et al., 2024), and LLaVA-Med (Li et al., 2023). Despite their potential, these models are susceptible to severe hallucinations that critically undermine diagnostic trustworthiness.

In clinical contexts, these errors manifest as fabricated lesions or exaggerated pathologies. Mitigation efforts prioritize specialized instruction tuning and uncertainty quantification to restore reliability. A notable advancement is MedHallTune (Yan et al., 2025), a dataset comprising over 100k images and 1M instruction pairs designed to contrast hallucinated and non-hallucinated responses. Fine-tuning on this curated data significantly improves grounding, boosting performance on standard benchmarks like VQA-RAD (Lau et al., 2018) and SLAKE (Liu et al., 2021). Alongside uncertainty-aware evaluation frameworks (Chen et al., 2024), these data-centric interventions are essential for ensuring the clinical reliability of medical VLMs.

- *Remote Sensing.* Remote sensing technology leverages high-altitude imagery to analyze terrestrial phenomena, facilitating applications in environmental monitoring, agriculture, and disaster management. The integration of I2T models has advanced capabilities in this domain, enabling tasks such as image captioning (Zhao et al., 2021), scene classification (Liu et al., 2024), and complex reasoning (Wang et al., 2024). However, the complexity of visual content and unique bird’s perspective in remote sensing images often lead I2T models to generate hallucinations. Generic models like BLIP-2 (Li et al., 2023) and MiniGPT-4 (Zhu et al., 2024) struggle significantly in this context, achieving less than 54% accuracy on the RSVQA benchmark due to persistent fabrication errors (Hu et al., 2025). To address these limitations, the community has developed specialized fine-tuning datasets (Wang et al., 2023; Du et al., 2019) and hallucination-focused benchmarks (Li et al., 2025). While fine-tuning on these domain-specific resources partially mitigates the issue, developing robust and reliable models for safety-critical remote sensing tasks remains an ongoing challenge.
- *Agriculture.* I2T models show promise in optimizing agricultural production through pests and diseases understanding (Truong et al., 2025; Li et al., 2025), soil analysis (Hong et al., 2024), and robotic navigation (Zhao et al., 2025; Zhou et al., 2024). Despite these capabilities, environmental distribution shifts remain a

primary driver of hallucination. The high variability in crop types, weather, and farming practices across regions creates a domain gap that degrades zero-shot performance. For instance, models frequently fail or hallucinate during leaf segmentation tasks when applied to unseen environments (Chawla et al., 2021). Addressing these failures requires robust domain adaptation strategies and localized data collection for model training and consistent performance across diverse agricultural settings.

Summary of MLLM Hallucinations in Key Domains. While MLLMs offer transformative potential in specialized domains, their deployment is impeded by critical safety risks. In high-stakes fields, model failures translate directly into real-world harms, such as physical dangers in embodied AI and misdiagnoses in medical imaging. We argue that generic mitigation strategies are insufficient for these contexts, necessitating the development of robust, domain-specific interventions Ojha et al. (2025); Noorani et al. (2025). To address this, we advocate for a sociotechnical framework that fuses expert-curated nuance, external verification, and human oversight. By prioritizing collaborative reliability over pure autonomy, this approach secures the essential trust required for deployment in safety-critical sectors.

4 Hallucination Detection Methods

In this chapter, we systematically review the existing hallucination detection methods for both I2T and T2I models. Tab. 3 summarizes the key details of the methods, including detection type, granularity, and technical approach. Fig. 5 summarizes three common hallucination detection methods for I2T and T2I models, including detector-based, caption-based and VQA-based methods. The specific I2T and T2I hallucination detection methods are presented in Sec. 4.1 and Sec. 4.2 respectively, and unified methods in Sec. 4.3, with analysis of their relationships in Sec. 4.4.

4.1 Hallucination Detection Methods for I2T Models

Hallucination detection is critical for assessing VLM reliability, particularly in I2T tasks like

image captioning and VQA. Depending on the accessibility of model internals, detection strategies can be broadly categorized into black-box and white-box approaches (Chakraborty et al., 2025). Given an input image, VLMs typically produce a sequence of output tokens, token-level probabilities, and internal layer representations. Black-box methods rely only on textual outputs, often using external evaluators or models, while white-box methods utilize internal signals such as attention weights or embeddings. This classification clarifies trade-offs between applicability and interpretability and reflects real-world deployment constraints. The following sections categorize existing I2T hallucination detection methods as black-box or white-box, highlighting their assumptions, mechanisms, and key applications.

4.1.1 Black-Box

Black-box methods lack access to VLM internals and rely on external tools for hallucination detection. They can be grouped into four types: external model-based, word matching, uncertainty-based, and similarity-based methods. External model-based methods use LLMs, VQA, or caption models to analyze VLM outputs. Word matching compares generated text with references or input content. Uncertainty-based detection identifies hallucinations via entropy or confidence scores, assuming hallucinations correlate with higher uncertainty. Similarity-based detection employs I2T models to generate reference images and compute image-image similarity.

VQA-based Detection. FaithScore (Jing et al., 2023) evaluates hallucinations in generative questions generated by VLMs. It first leverages GPT to identify descriptive content in the model’s response, then breaks down this content into fine-grained atomic facts. A visual entailment model (VEM) is used to verify the consistency of these atomic facts with the visual information, helping to detect hallucinations in the response. The work utilizes FaithScore as the evaluation metric. The function $s(e_j^i, I)$ represents the VEM, where an output of 1 indicates that the image covers the content of the atomic fact. The final hallucination score is calculated by weighting these binary scores, shown in Tab. 1.

Table 3 Summary of Hallucination Detection Methods in I2T, T2I, and Unified Scenarios

Task	Method	Access Type	Faithfulness Hallucination Detection Granularity			Detection Method
			Object-level	Scene-level	Attribute-level	
I2T	GAVIE (Liu et al., 2023)	Black-box	✓	✓	✓	LLM
	Throne (Kaul et al., 2024)	Black-box	✓	✗	✗	LLM
	HaELM (Wang et al., 2023)	Black-box	✓	✓	✓	LLM
	FaithScore (Jing et al., 2023)	Black-box	✓	✓	✓	VQA
	CutPaste & Find (Nguyen et al., 2025)	Black-box	✓	✓	✓	VQA
	FactCheXcker (Heiman et al., 2025)	Black-box	✓	✓	✗	VQA
	M-HalDetect (Gunjal et al., 2023)	Black-box	✓	✓	✓	Caption
	AI-Feedback (Xiao et al., 2025)	Black-box	✓	✓	✓	Caption
	HalLocalizer (Park et al., 2025)	Black-box	✓	✓	✓	Caption
	CHAIR (Rohrbach et al., 2018)	Black-box	✓	✗	✗	Word Matching
	NOPE (Lovenia et al., 2024)	Black-box	✓	✗	✗	Word Matching
	VL-Uncertainty (Zhang et al., 2024a)	Black-box	✓	✓	✓	Uncertainty
	Image2Text2Image (Huang et al., 2025)	Black-box	✓	✓	✓	Similarity
	DHCP (Zhang et al., 2024b)	White-box	✓	✗	✗	Attention
	OPERA (Huang et al., 2024)	White-box	✓	✗	✗	Attention
	HallE-Control (Zhai et al., 2023)	White-box	✓	✓	✓	Feature
	PROJECTAWAY (Jiang et al., 2025)	White-box	✓	✗	✗	Logits
	ContextualLens (Phukan et al., 2025)	White-box	✓	✓	✓	Logits
T2I	SR _{2D} & VISOR (Gokhale et al., 2022)	Black-box	✓	✓	✗	Detector
	DALL-Eval (Cho et al., 2022)	Black-box	✓	✗	✗	Detector
	VPEval (Cho et al., 2023)	Black-box	✓	✓	✗	Detector & VQA
	TIFA (Hu et al., 2023)	Black-box	✓	✓	✓	VQA
	T2I-CompBench (Huang et al., 2023)	Black-box	✓	✓	✓	VQA & Detector & Caption
	GraphQA (Qin et al., 2024)	Black-box	✓	✓	✓	VQA
	DSG (Cho et al., 2024)	Black-box	✓	✓	✓	VQA
	VQAScore (Lin et al., 2024)	Black-box	✓	✓	✓	VQA
	LLMScore (Lu et al., 2023)	Black-box	✓	✓	✓	Detector & Caption
Unified	UNIHD (Chen et al., 2024)	Black-box	✓	✗	✓	VQA
	CSN (Fei et al., 2024)	Black-box	✓	✓	✓	VQA

CutPaste & Find (Nguyen et al., 2025) is a training-free framework for detecting hallucinations in VLMs, leveraging a visual knowledge base and a VQA model. The knowledge base, built from Visual Genome (Krishna et al., 2016), contains object crops, attributes, and relationships, serving as a structured visual reference. The detection pipeline consists of three main steps: (1) extracting a scene graph from the model-generated caption; (2) locating the mentioned objects via object detection; and (3) verifying the consistency of the caption through image similarity queries and VQA-based checks. Hallucinations are identified when inconsistencies arise between the caption, detection results, and the visual knowledge base. Experiments on POPE and R-Bench show that CutPaste & Find achieves competitive performance compared to prior methods.

FactCheXcker (Heiman et al., 2025) proposes an extensible and modular pipeline to detect and mitigate object and measurement hallucinations in radiology reports generated by medical VLMs. It constructs a benchmark focused on EndoTracheal Tube (ETT) placement, leveraging the MIMIC-CXR dataset and multiple

report-generation models. The framework follows a query-code-update paradigm, where measurement queries are automatically generated from original reports, solved via domain-specific code modules, and incorporated back into updated reports. Evaluation metrics include precision for existence hallucination, Mean Absolute Error (MAE) for measurement hallucination, and composite scores combining both MAE and F1-score (shown in Tab. 1). Experiment results demonstrate that FactCheXcker significantly reduces hallucinations and improves both measurement accuracy and placement correctness across diverse models. Without requiring retraining, it serves as a valuable tool for enhancing clinical reliability in medical imaging AI systems.

LLM-based Detection. GAVIE (Liu et al., 2023) employs GPT to detect hallucinations in model responses. Specifically, the method inputs dense image descriptions with bounding box coordinates into GPT, then compares these descriptions to model responses for scoring. Evaluation criteria assess two dimensions: (1) Accuracy, measuring response consistency with image content; and (2) Relevance, determining whether responses

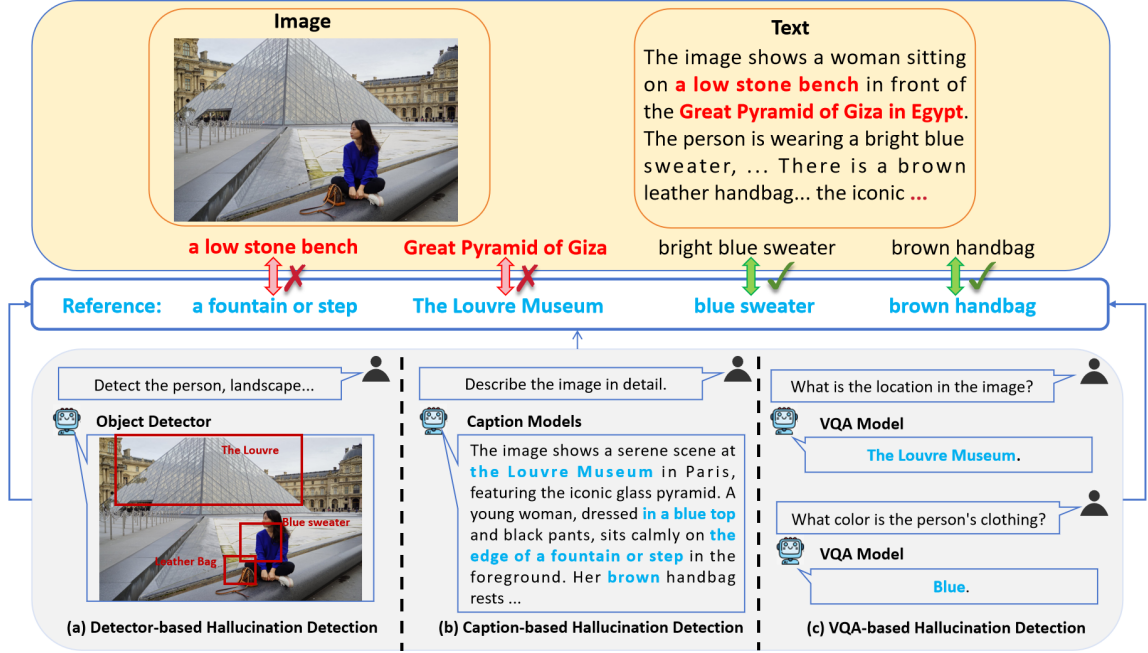


Fig. 5 Examples of detection methods. At the top is a mismatched image-text pair, and hallucinated text contents are highlighted in red. At the bottom are representative (a) Detector-based, (b) Caption-based, and (c) VQA-based hallucination detection methods arranged from left to right. The reference labels are highlighted in blue.

directly address instructions. Hallucination scores derived from this dual-aspect analysis identify hallucinated content. Similar strategies are adopted by CC-Eval (Zhai et al., 2023) and MMHal-Bench (Sun et al., 2024), which also rely on GPT to identify or score hallucinated contents.

Throne (Kaul et al., 2024) introduces an LLM-based framework for evaluating object-level hallucinations in free-form generative responses of VLMs. Throne employs public language models to analyze VLM-generated image descriptions prompted neutrally with “Describe this image in detail.” Through Abstractive Question Answering (AQA), it systematically queries the presence of each object class in ground-truth annotations, generating precision-recall metrics that directly quantify hallucination severity in generative tasks. Experiments show that compared to CHAIR (Rohrbach et al., 2018), Throne achieves lower hallucination misjudgment rates.

HaELM (Wang et al., 2023) uses a fine-tuned specialized LLM to serve as a hallucination detector for evaluating image descriptions. To construct the training corpus, it collects hallucination instances generated by VLMs and simulated via GPT models. The detector is based on

LLaMA and trained using this curated dataset. Experimental results show that HaELM achieves hallucination detection performance comparable to GPT-3.5. Moreover, HaELM can be deployed locally, offering advantages in reduced computational cost.

Caption-based Detection. M-HalDetect (Gunjal et al., 2023) introduces a hallucination detection framework based on a fine-tuned multi-modal reward model. It constructs a training set using approximately 4,000 MSCOCO (Lin et al., 2014) images paired with captions generated by InstructBLIP, annotated at the sub-sentence level to capture fine-grained hallucinations. The reward model is initialized from InstructBLIP by replacing its final embedding layer with a classification head, and is trained to predict hallucination labels at both the sentence and segment levels. To support flexible detection, the model is trained under two settings: binary (accurate vs. hallucinated) and ternary (accurate, hallucinated, and analytical). The ternary setup allows the model to distinguish factual errors from subjective analysis, reducing confusion in ambiguous cases. Experiments show that the model effectively detects

hallucinations in generated captions and generalizes well to outputs from other LVMs such as LLaVA and mPLUG-OWL.

AI-Feedback (Xiao et al., 2025) introduces a fine-grained hallucination detection framework for VLMs by constructing a detailed AI-generated feedback dataset. The model is trained to identify hallucinated content and assess its severity across various types, including object, attribute, and relationship. To build the dataset, 5000 images are sampled from Visual Genome (Krishna et al., 2016), and GPT-4 is leveraged to analyze the VLM-generated caption, which detects hallucinations and provides structured feedback. The detection model is trained on triplets consisting of the image, the VLM outputs, and the corresponding GPT feedback. In addition to enabling fine-grained hallucination detection, this work proposes Hallucination Severity-Aware Direct Preference Optimization (HSA-DPO), a mitigation strategy that leverages severity labels to guide preference-based fine-tuning.

HalLocalizer (Park et al., 2025) proposes a lightweight framework for token-level hallucination detection in the process of VLM generating text responses. Trained on the Hal-Loc dataset (150000 token-annotated samples across VQA, instruction-following, and captioning tasks), it provides probabilistic confidence scores rather than binary decisions, addressing ambiguity in real-world scenarios. The architecture offers dual input modalities: processing either the VLMs’ final hidden states or generated text through a VisualBERT encoder with four linear classification heads (for object/attribute/relationship/scene hallucination types). Designed for minimal computational overhead, HalLocalizer achieves efficient inference with low latency while generating per-token hallucination probabilities. Its plug-and-play integration preserves base model efficiency, enabling seamless deployment in existing VLMs to enhance reliability without significant resource demands.

Word Matching. CHAIR (Rohrbach et al., 2018) and NOPE (Lovenia et al., 2024) employ word matching paradigms for hallucination detection while implementing specialized rules for distinct hallucination types. CHAIR targets object-level hallucinations through a synonym mapping system for 80 MSCOCO (Lin et al., 2014) objects

(*e.g.*, “player”→“person”), involving text tokenization, singularization, and compound noun disambiguation (*e.g.*, distinguishing “hot dog” from “dog”). Its innovation lies in fusing dual annotation sources: segmentation labels verify object presence, while human captions accommodate referential variations (*e.g.*, semantic equivalence between “dining table” and “coffee table”). NOPE addresses negative semantic hallucinations via a dual evaluation framework. For Negative Pronoun (NegP) tasks, it applies rule-based accuracy to detect predefined negative pronouns ($A_{\text{NegP}} = \text{none, nothing, nobody}$); for non-negation tasks, it leverages METEOR metrics to assess unigram matches through weighted precision, recall, and alignment scores. Although relatively coarse, these methods offer simple and effective solutions; subsequent work such as OpenCHAIR (Ben-Kish et al., 2024) builds upon this foundation by employing large language models as advanced text aligners.

Uncertainty-based Detection. Zhang et al. (2024a) proposes VL-Uncertainty, a black-box hallucination detection method based on model uncertainty estimation. They observe that vision-language models tend to produce consistent outputs for semantically similar inputs when confident, but show variance when hallucinations occur. To measure this, it generates paraphrased queries and perturbed images (*e.g.*, Gaussian blur) and computes the entropy of model predictions across them. High entropy indicates uncertainty and potential hallucination. This method avoids manual annotations and demonstrates strong performance across 10 models and 4 benchmarks, highlighting uncertainty as a meaningful hallucination indicator.

Similarity-based Detection. Huang et al. (2025) introduces Image2Text2Image, a novel framework for evaluating the quality of image captioning based on image-to-image consistency, thereby removing the dependence on human-annotated references. It leverages diffusion models (*e.g.*, Stable Diffusion) to assess the fidelity of generated captions in a self-supervised manner. The proposed framework consists of four main components: (1) an image captioning module that generates textual descriptions from input images; (2) an image encoder that extracts visual features; (3) a diffusion model that synthesizes images

from the generated captions; and (4) a similarity module that compares the original and synthesized images in feature space. Low similarity scores suggest potential deficiencies in the captions, revealing discrepancies between the original visual content and its textual representation. This approach enables automatic evaluation without reliance on human labels.

4.1.2 White-Box

Unlike black-box methods, white-box approaches leverage internal representations for hallucination detection, providing detailed insights into the model’s generation process. Based on the accessed internal components, white-box detection can be categorized as attention-based, feature-based and logits-based methods. These approaches exploit distinguishable patterns between hallucinated and faithful content to detect as well as mitigate hallucinations effectively.

Attention-based Detection. DHCP (Zhang et al., 2024b) presents a white-box hallucination detection framework that exploits cross-modal attention patterns in LVLMs. Specifically, it captures the attention weights from the language model to visual tokens during the generation of the first output token, forming a structured (token $\times$ layer $\times$ head) attention tensor. Empirical analysis on the POPE-Extended dataset reveals that despite being textually identical, hallucinated and non-hallucinated responses exhibit significantly different attention distributions, which vary by hallucination type. Building on this observation, DHCP trains a lightweight two-stage MLP-based classifier (DHCP-d): the first-stage detector prioritizes high recall to flag potential hallucinations, while the second-stage refines predictions for higher precision. This cascaded design leads to a substantial performance improvement, achieving over 93% accuracy on the POPE-Extended test set. DHCP demonstrates that internal attention dynamics provide reliable and interpretable signals for hallucination detection without requiring access to model retraining or output references.

OPERA (Huang et al., 2024) detects and alleviates hallucination in VLMs by mitigating over-reliance on sparse summary tokens during decoding. Experiments show that hallucinations correlate with knowledge aggregation patterns in

self-attention, where models prioritize recent summary tokens over contextual image information. Based on this observation, OPERA introduces an over-trust penalty to suppress over-aggregation in beam-search logits and a retrospection-allocation strategy to trigger token re-selection when critical patterns emerge. Experiments show that OPERA generally improves the VLMs’ performances on popular benchmarks including POPE (Li et al., 2023) and MME (Fu et al., 2023).

Feature-based Detection. HallE-Control (Zhai et al., 2023) identifies that hallucinations occur when LVLM language responses contain finer-grained details than the visual module can verify, leading to unwarranted inferences. The method introduces a control module with a trainable projection layer. This layer transforms output embeddings in a way that reduces hallucinated content by modulating the model’s reliance on unverifiable details. Hallucinations are further reduced by adjusting control parameters to modulate knowledge integration. This parameter-efficient approach reduces object hallucination by 44% compared to LLaVA-7B while maintaining object coverage, enabling controllable generation of contextual versus parametric knowledge.

Logits-based Detection. PROJECTAWAY (Jiang et al., 2025) detects and alleviates object-level hallucination in VLMs through logits lens. This technique, originally introduced in the context of language models, involves directly mapping intermediate activations to the vocabulary space using the unembedding layer, allowing for an interpretable view of token predictions at different layers. By applying this technique to VLMs, PROJECTAWAY probes each image patch to determine the presence of objects. This approach reveals that real objects exhibit significantly higher output probabilities than hallucinated ones at the token level. These probability distributions further enable patch-level spatial localization of objects. Building on this granular analysis, the method applies knowledge erasure via linear orthogonalization of image features against hallucinated object features. Results show that targeted edits to a model’s latent representations achieve up to 25.7% hallucination reduction on MSCOCO (Lin et al., 2014) while preserving performance.

ContextualLens (Phukan et al., 2025) detects and mitigates hallucinations in VLMs through

token-level analysis and patch/bounding box-level grounding. Unlike the logit lens method that relies on non-contextual unembedding layer features (Jiang et al., 2025), ContextualLens leverages contextual token embeddings from middle VLM layers to capture multi-token concepts and spatial relationships. This enables granular hallucination detection across complex categories (attributes, actions, OCR) by computing cosine similarity between averaged answer token embeddings and image patch embeddings. For grounding, it generates precise bounding boxes via contextual embedding alignment, advancing zero-shot segmentation to grounded VQA. Evaluated on HQH benchmark (Yan et al., 2024), the method achieves significant improvements in mAP for hallucination detection while maintaining the training-free efficiency.

Summary of I2T Hallucination Detection. As shown in Tab. 3, hallucination detection primarily relies on black-box methods. Most black-box methods utilize external models to assist in detection, reducing manual labeling efforts. However, the detection results are constrained by the accuracy and generalizability of the auxiliary models (Li et al., 2024). To further improve the detection performance, some studies have constructed hallucination detection datasets and fine-tuned existing models as detectors for evaluation (Gunjal et al., 2023). However, these approaches incur high training costs, while uncertainty-based methods require no additional training but are prone to misjudging complex cases (Zhang et al., 2024a).

In comparison, white-box detection methods depend on the model’s internal signals, such as attention and feature. By accessing these internal representations, white-box methods enable the tracing of hallucination sources and offer valuable insights into the model’s generation process. They help reveal the flow of information and decision-making across layers, which can aid in diagnosing and mitigating specific hallucination cases. However, due to the complexity of internal mechanisms, most current white-box methods are generally limited to coarse-grained or holistic detection, struggling to distinguish fine-grained hallucination types including relationships (Xiao et al., 2025; Song et al., 2025).

Overall, while black-box methods tend to offer better generalizability and easier deployment, both approaches face significant limitations. Existing detection efforts are primarily centered on faithfulness hallucinations. The detection of factuality hallucinations, which require reasoning over external or commonsense knowledge, remains an underexplored challenge in the field.

Underexplored I2T Hallucination Detection Paradigms. While black-box methods are widely used for output analysis, existing white-box approaches leverage attention, feature, and logits signals (Zhai et al., 2023; Zhang et al., 2024b; Huang et al., 2024; Jiang et al., 2025; Phukan et al., 2025), which mostly focus on local tokens or single-step generation, lacking systematic monitoring of dynamic reasoning trajectories throughout the output sequence (Cafagna et al., 2023). Joint analysis of multi-modal internal signals remains limited, and there are few systematic metrics to quantify cross-modal inconsistencies. Consequently, most current detection methods are restricted to isolated indicators or specific hallucination types, leaving fine-grained detection of complex categories, semantic relations, and spatial grounding underexplored (Xiao et al., 2025; Song et al., 2025). Investigating these directions could enable more transparent, interpretable, and generalizable hallucination detection in I2T models.

4.2 Hallucination Detection Methods for T2I Models

Early assessments of image generation primarily focus on coarse-grained evaluations such as image quality, using metrics like FID (Heusel et al., 2017) and CLIPScore (Hessel et al., 2021). Building on these evaluations, hallucination detection in T2I models emphasizes image-text alignment at a fine-grained level, including object types, attributes, and relationships. Current hallucination detection approaches for T2I models largely rely on black-box methods, including object detection and VLMs. This section provides a comprehensive review of existing black-box methods for hallucination detection in T2I models.

4.2.1 Black-Box

Black-box hallucination detection methods in T2I tasks typically rely on external tools or models to verify whether the image accurately reflects the text prompt, including object detector, VQA model and caption model.

Detector-based Detection. SR_{2D} & VISOR (Gokhale et al., 2022) leverages pretrained object detectors and spatial relationship recognition algorithms to evaluate spatial consistency in T2I tasks. Objects and their spatial relationships are automatically identified in generated images. These elements are compared against the spatial layout specified in the text prompt to assess alignment with the textual description. This automated metric reveals severe spatial relationship limitations in T2I models, while demonstrating strong alignment with human judgment.

DALL-Eval (Cho et al., 2022) employs object detection models to quantify consistency failures in T2I models, specifically evaluating object existence, counting, and spatial relationships. The framework utilizes synthetic training data, which is derived from MSCOCO (Lin et al., 2014) and procedurally generated in Unity to train a DETR-based detector, which subsequently analyzes generated images for hallucinatory content.

VQA-based Detection. VPEval (Cho et al., 2023) employs a programmatic evaluation framework utilizing specialized visual modules to detect hallucinations across distinct categories, including object presence, OCR accuracy, spatial relationships, and quantitative counting. This approach dynamically assembles skill-specific evaluation pipelines, where dedicated modules (*e.g.*, object detection and counting systems) are invoked to verify prompt-image alignment for targeted attributes. As an interpretable visual programming paradigm, VPEval overcomes limitations of monolithic evaluation models by providing skill-optimized assessments with explanatory justifications, demonstrating significantly stronger correlation with human judgment than conventional single model-based evaluation.

TIFA (Hu et al., 2023) leverages LLMs and VQA models to assess T2I generation fidelity. For a given text prompt, question-answer pairs are automatically generated using an LLM and filtered by a question-answer model. These questions

are then posed to a VQA model using the generated image. Faithfulness is evaluated by assessing consistency between the VQA answers and the ground-truth answers derived from the text prompt. This reference-free metric enables fine-grained, interpretable evaluations, demonstrates superior correlation with human judgments, and reveals significant model deficiencies in spatial relations and object composition.

T2I-CompBench (Huang et al., 2023) evaluates compositional fidelity in synthesized images across three dimensions: object attribute, spatial relationships, and scene coherence. To assess object and attribute hallucinations, binary questions derived from text prompts are evaluated using VQA models. For spatial relationship verification, object detection combined with spatial recognition algorithms is employed. Regarding scene-level coherence, MiniGPT-4-generated descriptions are scored against textual requirements. Results show limitations of T2I models to generate complex scene compositions.

GraphQA (Qin et al., 2024) proposes a method for detecting hallucinations in T2I models by integrating scene graphs and VQA models. LLMs automatically generate evaluative questions from text prompts, while object detection algorithms (*e.g.*, GroundedSAM) and VLMs extract visual attributes and relationships to construct image knowledge graphs. Graph Question Answering then scores response consistency between the graphs and generated images to identify hallucinations. This structured approach establishes a quantitative, interpretable framework for evaluating image-text alignment that closely correlates with human assessment standards.

Davidsonian Scene Graph (DSG) (Cho et al., 2024) employs a graph-based framework with VQA to detect hallucinations in generated images. Unlike GraphQA’s approach (Qin et al., 2024), this methodology integrates dependency graphs to ensure comprehensive semantic coverage while preventing contradictory or redundant question generation. Drawing on linguistic formalization theories, the construction process recursively decomposes sentences into atomic propositions implemented as directed acyclic graphs. During VQA validation, negative responses to parent questions automatically terminate dependent sub-question evaluation. DSG guarantees question uniqueness and answer consistencies, thus

providing adaptable semantic coverage for robust image-text faithfulness evaluation.

VQAScore (Lin et al., 2024) leverages VQA architectures to quantify fine-grained text-visual alignment. The framework reformulates prompt-image consistency as a binary classification task, evaluating through VQA models with queries: “Does this figure show {Text}?”. Alignment scores derive from the probability of affirmative responses. To enhance detection capabilities, the method incorporates CLIP-FlanT5, a bidirectional image-question encoder fine-tuned specifically for hallucination identification. This approach establishes a state-of-the-art compositional evaluation metric that overcomes lexical ambiguity limitations in conventional methods, enabling reliable assessment of complex prompts across eight benchmarks.

Caption-based Detection. LLMScore (Lu et al., 2023) integrates object detection architectures with LLMs to identify hallucinations across object-level and scene-level granularity. Object detector localizes the visual entities, which is followed by region-to-text transformers and image captioning systems that generate localized and holistic descriptions. These descriptions are evaluated against the source text prompts via LLM-based alignment assessment to detect hallucinations. Results show that the framework demonstrates stronger correlation with human judgment than conventional metrics CLIP and BLIP.

Summary of T2I Hallucination Detection. Hallucination detection methods for T2I models mainly evaluate the consistency between generated images and textual prompts using external or fine-tuned models. Early works leverage specialized tools such as object detector (Gokhale et al., 2022). These methods are limited in coverage and mainly focus on coarse-granularity hallucination, such as object and spatial relationship. Another common approach is to use LLMs to decompose the prompt into relevant questions, and then employ VLMs to answer these questions based on the generated image (Qin et al., 2024; Cho et al., 2024), enabling the detection of various types of hallucinations. In addition, methods concerning factuality hallucinations are relatively few. Exploring comprehensive detection methods remains an important direction.

Underexplored T2I Hallucination Detection Paradigms. Current detection metrics are predominantly semantic, not physical. A model can generate a “car” that is semantically recognized by CLIP, TIFA, or VPEval, but the car might lack a shadow, have inconsistent reflections, or float slightly above the ground. This blindness to real-world constraints extends to cultural nuances, historical accuracy, and other factual contexts, where semantic correctness does not guarantee fidelity to reality (Borji, 2023). Moreover, critical internal issues, such as “catastrophic neglect” in attention heads (Chefer et al., 2023) and cross-modal misalignment during the latent denoising process (Hertz et al., 2023), remain largely unexamined by current output-based detection tools. Therefore, advancing the field requires a paradigm shift towards diagnostic methods that can probe the internal generation process. Future work should focus on methods that trace alignment during denoising, audit attention maps, and incorporate physical and cultural logic directly into the evaluation framework.

4.3 Unified Hallucination Detection Methods

Unified hallucination detection for I2T and T2I models is emerging as a promising direction for cross-modal evaluation. Some studies integrate both tasks via image-text alignment (Chen et al., 2024; Fei et al., 2024). Compared to task-specific methods, unified approaches capture both visual and textual content and typically require additional external models. Recent unified methods are mainly black-box, utilizing external tools and models to analyze image-text inconsistencies.

VQA-based Detection. UNIHD (Chen et al., 2024) introduces a unified multi-modal framework for detecting hallucinations in T2I and I2T tasks by assessing image-text consistency across four categories: object existence, attributes, scenes, and knowledge-based factuality. The methodology decomposes text prompts into atomic facts using GPT, and then generates verification questions from multiple perspectives. For visual validation, specialized tools, including object detectors, attribute classifiers (GPT-based), and external knowledge retrievers (*e.g.*, Google Search), operate in parallel to gather

evidentiary support. These outputs, combined with task metadata, are integrated into a VLM for hallucination assessment. This comprehensive approach establishes a novel paradigm for multi-category hallucination detection. It overcomes the limitations of task-specific methods through tool-assisted evidence validation and enables granular evaluation via the MHalubench benchmark.

CSN (Cross-graph Siamese Network) (Fei et al., 2024) establishes a unified hallucination detection framework for I2T and T2I tasks using structured scene graph representations. This approach detects object-existence, attribute, and relationship hallucinations through comparative analysis of dual scene graphs. Initially, a scene graph parser constructs a Visual Scene Graph (VSG) from generated images and a Textual Scene Graph (TSG) from input prompts. Subsequently, a pretrained Cross Graph Siamese Network with cross-attention mechanisms identifies semantic discrepancies between VSG and TSG representations. Ultimately, a large language model adjudicates hallucination occurrences based on detected inconsistencies. This structured framework pioneers fine-grained multi-modal hallucination detection by resolving semantic gaps in comprehension and generation processes, outperforming baselines on MHalubench.

Summary of Unified Hallucination Detection. Unified hallucination detection methods aim to provide a cross-modal framework for identifying hallucinations in both I2T and T2I tasks, often leveraging question decomposition, and graph-based reasoning, assisted by external tools and models. Existing challenges include dependence on external models, modality-specific hallucination patterns, and limited efficiency. Developing robust and efficient frameworks is a key direction for future research.

4.4 Relationship between Hallucination Detection in I2T and T2I Models

As shown in Tab. 3 and analysis above, hallucination detection in I2T and T2I models shares the common goal of identifying semantic inconsistencies between the input and output modalities (Chen et al., 2024; Fei et al., 2024). Both approaches commonly depend on external models such as VQA models (*e.g.*, CutPaste & Find,

TIFA) (Hu et al., 2023; Nguyen et al., 2025) or large language models (*e.g.*, GAVIE, LLM-Score) (Liu et al., 2023; Lu et al., 2023) to verify cross-modal consistency.

In addition to these shared characteristics, a difference lies in white-box methods. I2T models take advantage of techniques such as attention analysis (Zhang et al., 2024b) and feature (Zhai et al., 2023), which enable direct inspection of the generation process and help trace the origins of hallucinated content. T2I hallucination detection primarily relies on external tools, including object detectors and VQA models, and typically lacks direct access to internal generation signals. Future work could consider incorporating white-box strategies into T2I models to enhance detection effectiveness.

Current unified hallucination detection methods typically check the alignment between image and text content by combining multi-modal features (Chen et al., 2024; Fei et al., 2024). These methods combine outputs from multiple external pretrained models, allowing for fine-grained identification of hallucinations. Future research could explore new mechanisms, such as fine-grained uncertainty estimation, building effective unified frameworks that better capture hallucinations in both T2I and I2T tasks.

5 Challenges and Future Directions

Despite growing efforts and progress in MLLM hallucination evaluation, several critical challenges remain unresolved and underexplored. Below, we summarize these key issues to offer guidance and suggestions for future research.

5.1 Scalable, Rigorous and Comprehensive Hallucination Evaluation

The rapid advancement of MLLM reasoning capabilities has significantly increased the complexity of generated outputs (Mu et al., 2023; Shao et al., 2024; Zhao et al., 2025). This evolution underscores the urgent need for more sophisticated automatic evaluation methods that transcend simple binary or surface-level assessments of hallucination (Jiang et al., 2024; Jing et al., 2023; Rawte

et al., 2025; Zhou et al., 2024). However, existing hallucination benchmarks are diversifying in task format and difficulty, and their evaluations focus on model outputs in isolation, overlooking the role of training data quality, architectural choices, and pre-training objectives in shaping hallucination behavior (Kang et al., 2025). As a result, the understanding of hallucinations remains partial, context-dependent, and limited in scope. Therefore, the evolution of evaluation is supposed to transition from static, outcome-based to dynamic, process-oriented paradigms capable of distinguishing critical factual errors and identifying root causes from a full-cycle, comprehensive perspective.

5.2 Explainable and Reliable Hallucination Evaluation

Hallucinations in VLMs often result from subtle misalignments between visual and textual modalities, involving complex, non-linear, and context-dependent interactions (Rudin, 2019; Yang et al., 2023). Consequently, developing explainability methods that can disentangle and trace these multi-modal reasoning processes remains a significant challenge (Yan et al., 2024; Ben Melech Stan et al., 2024). As mentioned in Sec. 3.1 and Sec. 3.2, existing hallucination evaluation benchmarks predominantly rely on coarse-grained evaluation metrics, such as accuracy, which oversimplify the problem by ignoring the nuanced spectrum of hallucination types and severities. Moreover, as benchmarks built from known datasets, it will inadvertently overlap training data, giving an optimistic view of performance (Tu et al., 2025). This lack of interpretability and reliability limits insight into the origins and locations of hallucinations, impeding effective diagnosis and mitigation. Advancing standardizing benchmark formats and explainable evaluation is therefore critical for enabling root cause analysis and allow more meaningful model comparisons.

5.3 Domain-specific Factuality Hallucination Evaluation

Evaluating factuality hallucinations in MLLMs poses a particular challenge when applied to specialized domains such as medicine (Ayaz et al., 2024) and embodied intelligence (Wu et al., 2025;

Sarch et al., 2024). Unlike general-domain scenarios, these fields demand strict adherence to complex, technical, and often regulated knowledge. However, efforts to systematically evaluate hallucinations in these domain-specific contexts remain limited. As Tab. 2 shows, while benchmarks like Med-HallMark (Chen et al., 2024) target medical VLMs, comparable frameworks for other specialized disciplines are largely absent. Future frameworks are expected to prioritize generalization-oriented environmental settings to rigorously assess model robustness against the unpredictable nature of practical deployment and guarantee deployment readiness in safety-critical domains.

5.4 Unified Detection for I2T and T2I Models

Current research on I2T and T2I hallucination detection faces notable gaps, including limited investigation and fragmented evaluation standards. However, the landscape is shifting towards unified paradigms that address hallucination analysis holistically across both modalities. Pioneering approaches like UNIHD (Chen et al., 2024) demonstrate the efficacy of consistency-based paradigms that generalize across I2T and T2I tasks. Modular architectures such as VPEval (Cho et al., 2023) employ specialized submodules to localize hallucinations at a fine-grained level. These methods, however, typically depend on external tools (*e.g.*, object detectors or knowledge retrievers) for content validation, which can complicate deployment and scalability. To advance the field, future research should prioritize the development of standardized, extensible frameworks that support unified detection across modalities. Establishing these foundations will be critical for improving reproducibility, fostering cross-modal insights, and enabling more robust, interpretable multimodal systems.

5.5 Hallucination Evaluation in Real-World Scenarios

While existing benchmarks such as POPE (Li et al., 2023), MME (Fu et al., 2023), HallusionBench (Guan et al., 2023), and Med-HallMark (Chen et al., 2024) offer precise measurements through well-defined ground truths,

they predominantly foster an “exam-oriented” evaluation paradigm (Kalai et al., 2025). This reliance on curated or synthetic datasets creates a significant virtual-to-real discrepancy: models may demonstrate high proficiency in controlled settings (*i.e.* overfitting dataset) yet encounter failures when facing the ambiguous, long-tail distributions inherent to real-world applications. To bridge this gap, MLLM evaluation should evolve beyond static dataset performance and draw on advancements in the LLM domain, such as Reeval’s (Yu et al., 2024) use of adversarial perturbations and HaluEval-Wild’s (Zhu et al., 2024) focus on real user interactions. To ensure model reliability, the paradigm must shift from evaluating static accuracy to assessing interactive robustness. This change is indispensable for establishing a continuous feedback loop that not only identifies current errors but allows for the proactive prediction of failure modes in novel, safety-critical environments.

6 Conclusion

Recent MLLMs have demonstrated strong capabilities in multi-modal tasks, but still suffer from the hallucination problem. This survey provides a comprehensive review of hallucination issues observed in existing MLLMs, with a particular focus on both I2T and T2I paradigms. We systematically categorize hallucinations into faithfulness and factuality types and analyze current evaluation methodologies. We highlight several key trends in the field, including the emergence of automated benchmark construction, the movement towards fine-grained evaluation, and the shift towards a unified perspective that bridges evaluation approaches between I2T and T2I tasks. Furthermore, our review of hallucination detection methods highlights the growing reliance on external tools to enable black-box detection strategies. Based on these insights, we outline several promising future directions for advancing hallucination evaluation in MLLMs. We hope this survey could provide researchers in this field with a clearer understanding of the current landscape and offer valuable guidance for future exploration.

Data Availability Statement

Data availability is not applicable to this paper as no new data were created or analyzed.

References

- Open AI: Hello GPT-4o. <https://openai.com/zh-Hans-CN/index/hello-gpt-4o/>. Accessed: 2025-07-13 (2024)
- Team, G., Anil, R., Borgeaud, S., Alayrac, J.-B., Yu, J., Soricut, R., Schalkwyk, J., Dai, A.M., Hauth, A., Millican, K., et al.: Gemini: a family of highly capable multimodal models. *CoRR* (2023) [2312.11805](#)
- Wang, P., Bai, S., Tan, S., Wang, S., Fan, Z., Bai, J., Chen, K., Liu, X., Wang, J., Ge, W., et al.: Qwen2-vl: Enhancing vision-language model’s perception of the world at any resolution. *CoRR* (2024) [2409.12191](#)
- Rombach, R., Blattmann, A., Lorenz, D., Esser, P., Ommer, B.: High-resolution image synthesis with latent diffusion models. *CVPR*, 10684–10695 (2022)
- Ramesh, A., Pavlov, M., Goh, G., Gray, S., Voss, C., Radford, A., Chen, M., Sutskever, I.: Zero-shot text-to-image generation. *ICML*, 8821–8831 (2021). Pmlr
- Zhang, H., Shao, W., Liu, H., Ma, Y., Luo, P., Qiao, Y., Zheng, N., Zhang, K.: B-avibench: Towards evaluating the robustness of large vision-language model on black-box adversarial visual-instructions. *TIFS* (2024)
- Guo, Q., Pang, S., Jia, X., Liu, Y., Guo, Q.: Efficient generation of targeted and transferable adversarial examples for vision-language models via diffusion models. *TIFS* (2024)
- Aafaq, N., Akhtar, N., Liu, W., Shah, M., Mian, A.: Language model agnostic gray-box adversarial attack on image captioning. *TIFS* **18**, 626–638 (2022)
- Zhang, H., Tang, H., Sun, Y., He, S., Li, Z.: Modality-specific interactive attack for vision-language pre-training models. *TIFS* (2025)
- Fan, W., Li, H., Jiang, W., Hao, M., Yu, S., Zhang, X.: Stealthy targeted backdoor attacks against image captioning. *TIFS* **19**, 5655–5667 (2024)
- Huang, K., Sun, K., Xie, E., Li, Z., Liu, X.: T2i-compbench: A comprehensive benchmark for open-world compositional text-to-image generation. *NeurIPS* **36**, 78723–78747 (2023)
- Bai, Z., Wang, P., Xiao, T., He, T., Han, Z., Zhang, Z., Shou, M.Z.: Hallucination of multimodal large language models: A survey. *CoRR* (2024) [2404.18930](#)
- Lan, W., Chen, W., Chen, Q., Pan, S., Zhou, H., Pan, Y.: A survey of hallucination in large visual language models. *CoRR* (2024) [2410.15359](#)

- Liu, H., Xue, W., Chen, Y., Chen, D., Zhao, X., Wang, K., Hou, L., Li, R., Peng, W.: A survey on hallucination in large vision-language models. *CoRR* (2024) [2402.00253](#)
- Rohrbach, A., Hendricks, L.A., Burns, K., Darrell, T., Saenko, K.: Object hallucination in image captioning. *EMNLP*, 4035–4045 (2018)
- Li, Y., Du, Y., Zhou, K., Wang, J., Zhao, W.X., Wen, J.-r.: Evaluating object hallucination in large vision-language models. *EMNLP* (2023)
- Chen, X., Wang, C., Xue, Y., Zhang, N., Yang, X., Li, Q., Shen, Y., Liang, L., Gu, J., Chen, H.: Unified hallucination detection for multimodal large language models. *ACL* (2024)
- Hu, Y., Liu, B., Kasai, J., Wang, Y., Ostendorf, M., Krishna, R., Smith, N.A.: Tifa: Accurate and interpretable text-to-image faithfulness evaluation with question answering. *ICCV*, 20349–20360 (2023)
- Wu, M.-K., Ji, J., Huang, O., Li, J., Wu, Y., Sun, X., Ji, R.: Evaluating and analyzing relationship hallucinations in large vision-language models. *ICML* (2024)
- Fu, C., Chen, P., Shen, Y., Qin, Y., Zhang, M., Lin, X., Yang, J., Zheng, X., Li, K., Sun, X., et al.: Mme: A comprehensive evaluation benchmark for multimodal large language models. *CoRR* (2023) [2306.13394](#)
- Seth, A., Manocha, D., Agarwal, C.: Hallucinogen: A benchmark for evaluating object hallucination in large visual-language models. *CoRR* (2024) [2412.20622](#)
- Zhou, K., Zhu, Y., Chen, Z., Chen, W., Zhao, W.X., Chen, X., Lin, Y., Wen, J.-R., Han, J.: Don’t make your llm an evaluation benchmark cheater. *CoRR* (2023) [2311.01964](#)
- Feng, W., He, X., Fu, T.-J., Jampani, V., Akula, A., Narayana, P., Basu, S., Wang, X.E., Wang, W.Y.: Training-free structured diffusion guidance for compositional text-to-image synthesis. *ICLR* (2023)
- Gokhale, T., Palangi, H., Nushi, B., Vineet, V., Horvitz, E., Kamar, E., Baral, C., Yang, Y.: Benchmarking spatial relationships in text-to-image generation. *CoRR* (2022) [2212.10015](#)
- Huang, Z., He, W., Long, Q., Wang, Y., Li, H., Yu, Z., Shu, F., Chan, L., Jiang, H., Gan, L., et al.: T2i-factualbench: Benchmarking the factuality of text-to-image models with knowledge-intensive concepts. *CoRR* (2024) [2412.04300](#)
- Danish, S., Sadeghi-Niaraki, A., Khan, S.U., Dang, L.M., Tightiz, L., Moon, H.: A comprehensive survey of vision-language models: Pretrained models, fine-tuning, prompt engineering, adapters, and benchmark datasets. *Information Fusion*, 103623 (2025)
- Shen, H., Zhang, J., Xiong, B., Hu, R., Chen, S., Wan, Z., Wang, X., Zhang, Y., Gong, Z., Bao, G., et al.: Efficient diffusion models: A survey. *TMLR* (2025)
- Ma, Z., Zhang, Y., Jia, G., Zhao, L., Ma, Y., Ma, M., Liu, G., Zhang, K., Ding, N., Li, J., et al.: Efficient diffusion models: A comprehensive survey from principles to practices. *IEEE TPAMI* (2025)
- Chen, X., Ma, Z., Zhang, X., Xu, S., Qian, S., Yang, J., Fouhey, D., Chai, J.: Multi-object hallucination in vision language models. *NeurIPS* **37**, 44393–44418 (2024)
- Huang, W., Liu, H., Guo, M., Gong, N.: Visual hallucinations of multi-modal large language models. *Findings of the ACL*, 9614–9631 (2024)
- Yan, B., Chen, Z., Min, Y., Zhang, J., Wang, J., Wang, X., Shan, S.: Shale: A scalable benchmark for fine-grained hallucination evaluation in vlms. *ACM MM*, 13442–13449 (2025)
- Park, E., Kim, M., Kim, G.: Halloc: Token-level localization of hallucinations for vision language models. *CVPR*, 29893–29903 (2025)
- Bakr, E.M., Sun, P., Shen, X., Khan, F.F., Li, L.E., Elhoseiny, M.: Hrs-bench: Holistic, reliable and scalable benchmark for text-to-image models. *ICCV*, 20041–20053 (2023)
- Jiang, C., Ye, W., Dong, M., Jia, H., Xu, H., Yan, M., Zhang, J., Zhang, S.: Hal-eval: A universal and fine-grained hallucination evaluation framework for large vision language models. *ACM MM* (2024)
- Ayaz, M., Khan, M., Saqib, M., Khelifi, A., Sajjad, M., Elsaddik, A.: Medvlm: Medical vision-language model for consumer devices. *IEEE CEM* (2024)
- Arshad, M.A., Jubery, T.Z., Roy, T., Nassiri, R., Singh, A.K., Singh, A., Hegde, C., Ganapathysubramanian, B., Balu, A., Krishnamurthy, A., et al.: Leveraging vision language models for specialized agricultural tasks. *WACV*, 6320–6329 (2025). IEEE
- Lim, Y., Choi, H., Shim, H.: Evaluating image hallucination in text-to-image generation with question-answering. *AAAI* **39**(25), 26290–26298 (2025)
- Hu, H., Zhang, J., Zhao, M., Sun, Z.: Ciem: Contrastive instruction evaluation method for better instruction tuning. *NeurIPS Workshop* (2023)
- Chen, Z., Zhu, Y., Zhan, Y., Li, Z., Zhao, C., Wang, J., Tang, M.: Mitigating hallucination in visual language models with visual supervision. *CoRR* (2023) [2311.16479](#)
- Guan, T., Liu, F., Wu, X., Xian, R., Li, Z., Liu, X., Wang, X., Chen, L., Huang, F., Yacoob, Y., Manocha, D., Zhou, T.: Hallusionbench: An advanced diagnostic suite for entangled language hallucination and visual illusion in large vision-language models. *CVPR*, 14375–14385 (2023)
- Wang, L., He, J., Li, S., Liu, N., Lim, E.-P.: Mitigating fine-grained hallucination by fine-tuning

- large vision-language models with caption rewrites. *MMM* (2023)
- Qiu, H., Huang, J., Gao, P., Qi, Q., Zhang, X., Shao, L., Lu, S.: Longhalqa: Long-context hallucination evaluation for multimodal large language models. *CoRR* (2024) [2410.09962](#)
- Liu, J., Fu, Y., Xie, R., Xie, R., Sun, X., Lian, F., Kang, Z., Li, X.: Phd: A chatgpt-prompted visual hallucination evaluation dataset. *CVPR*, 19857–19866 (2025)
- Cao, Q., Cheng, J., Liang, X., Lin, L.: Visdiahallbench: A visual dialogue benchmark for diagnosing hallucination in large vision-language models. *ACL*, 12161–12176 (2024)
- Wu, X., Guan, T., Li, D., Huang, S., Liu, X., Wang, X., Xian, R., Shrivastava, A., Huang, F., Boyd-Graber, J., et al.: Autohallusion: Automatic generation of hallucination benchmarks for vision-language models. *Findings of the EMNLP*, 8395–8419 (2024)
- Ben-Kish, A., Yanuka, M., Alper, M., Giryas, R., Averbuch-Elor, H.: Mitigating open-vocabulary caption hallucinations. *EMNLP*, 22680–22698 (2024)
- Zhai, B., Yang, S., Xu, C., Shen, S., Keutzer, K., Li, C., Li, M.: Halle-control: controlling object hallucination in large multimodal models. *CoRR* (2023) [2310.01779](#)
- Sun, Z., Shen, S., Cao, S., Liu, H., Li, C., Shen, Y., Gan, C., Gui, L., Wang, Y.-X., Yang, Y., et al.: Aligning large multimodal models with factually augmented rlhf. *Findings of the ACL*, 13088–13110 (2024)
- Liu, F., Lin, K., Li, L., Wang, J., Yacoob, Y., Wang, L.: Mitigating hallucination in large multi-modal models via robust instruction tuning. *ICLR* (2023)
- Lovenia, H., Dai, W., Cahyawijaya, S., Ji, Z., Fung, P.: Negative object presence evaluation (nope) to measure object hallucination in vision-language models. *ALVR Workshops*, 37–58 (2024)
- Wang, J., Zhou, Y., Xu, G., Shi, P., Zhao, C., Xu, H., Ye, Q., Yan, M., Zhang, J., Zhu, J., et al.: Evaluation and analysis of hallucination in large vision-language models. *CoRR* (2023) [2308.15126](#)
- Rani, A., Rawte, V., Sharma, H., Anand, N., Rajbangshi, K., Sheth, A., Das, A.: Visual hallucination: Definition, quantification, and prescriptive remediations. *CoRR* (2024) [2403.17306](#)
- Villa, A., Léon, J., Soto, A., Ghanem, B.: Behind the magic, merlim: Multi-modal evaluation benchmark for large image-language models. *CVPR*, 492–502 (2025)
- Wang, J., Wang, Y., Xu, G., Zhang, J., Gu, Y., Jia, H., Wang, J., Xu, H., Yan, M., Zhang, J., et al.: Amber: An llm-free multi-dimensional benchmark for mllms hallucination evaluation. *CoRR* (2023) [2311.07397](#)
- Chen, J., Yang, D., Wu, T., Jiang, Y., Hou, X., Li, M., Wang, S., Xiao, D., Li, K., Zhang, L.: Detecting and evaluating medical hallucinations in large vision language models. *CoRR* (2024) [2406.10185](#)
- Tu, Y., Hu, R., Sang, J.: Ode: Open-set evaluation of hallucinations in multimodal large language models. *CVPR*, 19836–19845 (2025)
- Cho, J., Zala, A., Bansal, M.: Dall-eval: Probing the reasoning skills and social biases of text-to-image generation models. *ICCV*, 3043–3054 (2023)
- Niu, Y., Ning, M., Zheng, M., Lin, B., Jin, P., Liao, J., Ning, K., Zhu, B., Yuan, L.: Wise: A world knowledge-informed semantic evaluation for text-to-image generation. *CoRR* (2025) [2503.07265](#)
- Li, B., Lin, Z., Pathak, D., Li, J., Fei, Y., Wu, K., Xia, X., Zhang, P., Neubig, G., Ramanan, D.: Evaluating and improving compositional text-to-visual generation. *CVPR*, 5290–5301 (2024)
- Jing, L., Li, R., Chen, Y., Du, X.: Faithscore: Fine-grained evaluations of hallucinations in large vision-language models. *EMNLP* (2023)
- Kaul, P., Li, Z., Yang, H., Dukler, Y., Swaminathan, A., Taylor, C., Soatto, S.: Throne: An object-based hallucination benchmark for the free-form generations of large vision-language models. *CVPR*, 27228–27238 (2024)
- Nguyen, C.-D., Wu, X., Vu, D.A., Zhao, S., Nguyen, T., Luu, A.T.: Cutpaste&find: Efficient multimodal hallucination detector with visual-aid knowledge base. *CoRR* (2025) [2502.12591](#)
- Huang, J.-H., Zhu, H., Shen, Y., Rudinac, S., Kanoulas, E.: Image2text2image: A novel framework for label-free evaluation of image-to-text generation with text-to-image diffusion models. *MMM*, 413–427 (2025). Springer
- Gunjal, A., Yin, J., Bas, E.: Detecting and preventing hallucinations in large vision language models. *AAAI* (2023)
- Xiao, W., Huang, Z., Gan, L., He, W., Li, H., Yu, Z., Shu, F., Jiang, H., Zhu, L.: Detecting and mitigating hallucination in large vision language models via fine-grained ai feedback. *AAAI* (2025)
- Heiman, A., Zhang, X., Chen, E., Kim, S.E., Rajpurkar, P.: Factchexcker: Mitigating measurement hallucinations in chest x-ray report generation models. *CVPR*, 30787–30796 (2025)
- Zhang, R., Zhang, H., Zheng, Z.: Vl-uncertainty: Detecting hallucination in large vision-language model via uncertainty estimation. *CoRR* (2024) [2411.11919](#)
- Zhang, Y., Xie, R., Chen, J., Sun, X., Wang, Y., et al.: Dhcp: Detecting hallucinations by cross-modal attention pattern in large vision-language models. *CoRR* (2024) [2411.18659](#)
- Huang, Q., Dong, X., Zhang, P., Wang, B., He, C.,

- Wang, J., Lin, D., Zhang, W., Yu, N.: Opera: Alleviating hallucination in multi-modal large language models via over-trust penalty and retrospection-allocation. *CVPR*, 13418–13427 (2024)
- Jiang, N., Kachinthaya, A., Petryk, S., Gandelsman, Y.: Interpreting and editing vision-language representations to mitigate hallucinations. *ICLR* (2025)
- Phukan, A., Divyansh, D., Morj, H.K., Vaishnavi, V., Saxena, A., Goswami, K.: Beyond logit lens: Contextual embeddings for robust hallucination detection & grounding in vlms. *NAACL*, 9661–9675 (2025)
- Cho, J., Zala, A., Bansal, M.: Dall-eval: Probing the reasoning skills and social biases of text-to-image generation models. *ICCV*, 3020–3031 (2022)
- Cho, J., Zala, A., Bansal, M.: Visual programming for step-by-step text-to-image generation and evaluation. *NeurIPS* **36**, 6048–6069 (2023)
- Lu, Y., Yang, X., Li, X., Wang, X.E., Wang, W.Y.: Llmscore: Unveiling the power of large language models in text-to-image synthesis evaluation. *NeurIPS* **36**, 23075–23093 (2023)
- Qin, Z., Cheng, D., Wang, H., Yi, H., Shao, Y., Fan, Z., Li, K., Lao, Q.: Evaluating hallucination in text-to-image diffusion models with scene-graph based question-answering agent. *CoRR* (2024) [2412.05722](https://arxiv.org/abs/2412.05722)
- Cho, J., Hu, Y., Baldrige, J.M., Garg, R., Anderson, P., Krishna, R., Bansal, M., Pont-Tuset, J., Wang, S.: Davidsonian scene graph: Improving reliability in fine-grained evaluation for text-to-image generation. *ICLR* (2024)
- Lin, Z., Pathak, D., Li, B., Li, J., Xia, X., Neubig, G., Zhang, P., Ramanan, D.: Evaluating text-to-visual generation with image-to-text generation. *ECCV*, 366–384 (2024). Springer
- Fei, H., Luo, M., Xu, J., Wu, S., Ji, W., Lee, M.-L., Hsu, W.: Fine-grained structural hallucination detection for unified visual comprehension and generation in multimodal llm. *ACM MM Workshops*, 13–22 (2024)
- Liu, H., Li, C., Wu, Q., Lee, Y.J.: Visual instruction tuning. *NeurIPS* **36**, 34892–34916 (2023)
- OpenAI: GPT-4V(ision) System Card. https://cdn.openai.com/papers/GPTV_System_Card.pdf. Accessed: 2025-12-04 (2023)
- Radford, A., Kim, J.W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., Krueger, G., Sutskever, I.: Learning transferable visual models from natural language supervision. *ICML* (2021)
- Liu, H., Li, C., Li, Y., Li, B., Zhang, Y., Shen, S., Lee, Y.J.: Llava-next: Improved reasoning, ocr, and world knowledge, january 2024. URL [https://llava-vl.github.io/blog/2024-01-30-llava-next-1\(8\)](https://llava-vl.github.io/blog/2024-01-30-llava-next-1(8)) (2024)
- Liu, N.F., Lin, K., Hewitt, J., Paranjape, A., Bevilacqua, M., Petroni, F., Liang, P.: Lost in the middle: How language models use long contexts. *ACL* **12**, 157–173 (2024)
- Kang, S., Kim, J., Kim, J., Hwang, S.J.: See what you are told: Visual attention sink in large multimodal models. *ICLR* (2025)
- Liang, V.W., Zhang, Y., Kwon, Y., Yeung, S., Zou, J.Y.: Mind the gap: Understanding the modality gap in multi-modal contrastive representation learning. *NeurIPS* **35**, 17612–17625 (2022)
- Li, J., Li, D., Savarese, S., Hoi, S.: Blip-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. *ICML*, 19730–19742 (2023)
- Yan, S., Bai, M., Chen, W., Zhou, X., Huang, Q., Li, L.E.: Vigor: Improving visual grounding of large vision language models with fine-grained reward modeling. *ECCV*, 37–53 (2024). Springer
- Alayrac, J.-B., Donahue, J., Luc, P., Miech, A., Barr, I., Hasson, Y., Lenc, K., Mensch, A., Millican, K., Reynolds, M., et al.: Flamingo: a visual language model for few-shot learning. *NeurIPS* **35**, 23716–23736 (2022)
- He, J., Zhu, K., Guo, H., Fang, J., Hua, Z., Jia, Y., Tang, M., Chua, T.-S., Wang, J.: Cracking the code of hallucination in vlms with vision-aware head divergence. *ACL*, 3488–3501 (2025)
- Pi, R., Miao, K., Peihang, L., Liu, R., Gao, J., Zhang, J., Zhou, X.: Pointing to a llama and call it a camel: On the sycophancy of multimodal large language models. *EMNLP*, 20177–20191 (2025)
- Zhu, T., Liu, Q., Wang, F., Tu, Z., Chen, M.: Unraveling cross-modality knowledge conflicts in large vision-language models. *CoRR* (2024) [2410.03659](https://arxiv.org/abs/2410.03659)
- Zhang, R., Han, J., Liu, C., Gao, P., Zhou, A., Hu, X., Yan, S., Lu, P., Li, H., Qiao, Y.: Llama-adapter: Efficient fine-tuning of language models with zero-init attention. *CoRR* (2023) [2303.16199](https://arxiv.org/abs/2303.16199)
- Liu, H., Li, C., Li, Y., Lee, Y.J.: Improved baselines with visual instruction tuning. *CVPR*, 26296–26306 (2024)
- Zhu, D., Chen, J., Shen, X., Li, X., Elhoseiny, M.: Minigpt-4: Enhancing vision-language understanding with advanced large language models. *ICLR* (2024)
- Christiano, P.F., Leike, J., Brown, T., Martic, M., Legg, S., Amodei, D.: Deep reinforcement learning from human preferences. *NeurIPS* **30** (2017)
- Rafailov, R., Sharma, A., Mitchell, E., Manning, C.D., Ermon, S., Finn, C.: Direct preference optimization: Your language model is secretly a reward model. *NeurIPS* **36**, 53728–53741 (2023)

- Zhao, Z., Wang, B., Ouyang, L., Dong, X., Wang, J., He, C.: Beyond hallucinations: Enhancing lvlms through hallucination-aware direct preference optimization. *CoRR* (2023) [2311.16839](#)
- Esser, P., Kulal, S., Blattmann, A., Entezari, R., Müller, J., Saini, H., Levi, Y., Lorenz, D., Sauer, A., Boesel, F., et al.: Scaling rectified flow transformers for high-resolution image synthesis. *ICML*, 12606–12633 (2024)
- Betker, J., Goh, G., Jing, L., Brooks, T., Wang, J., Li, L., Ouyang, L., Zhuang, J., Lee, J., Guo, Y., et al.: Improving image generation with better captions. Computer Science. <https://cdn.openai.com/papers/dall-e-3.pdf> **2**(3), 8 (2023)
- Peebles, W., Xie, S.: Scalable diffusion models with transformers. *CVPR*, 4195–4205 (2023)
- Koishigarina, D., Uselis, A., Oh, S.J.: Clip behaves like a bag-of-words model cross-modally but not unimodally. *CoRR* (2025) [2502.03566](#)
- Chung, H.W., Hou, L., Longpre, S., Zoph, B., Tay, Y., Fedus, W., Li, Y., Wang, X., Dehghani, M., Brahma, S., et al.: Scaling instruction-finetuned language models. *JMLR* **25**(70), 1–53 (2024)
- Touvron, H., Martin, L., Stone, K., Albert, P., Almahairi, A., Babaei, Y., Bashlykov, N., Batra, S., Bhargava, P., Bhosale, S., et al.: Llama 2: Open foundation and fine-tuned chat models. *CoRR* (2023) [2307.09288](#)
- Yang, A., Xiao, B., Wang, B., Zhang, B., Bian, C., Yin, C., Lv, C., Pan, D., Wang, D., Yan, D., et al.: Baichuan 2: Open large-scale language models. *CoRR* (2023) [2309.10305](#)
- Jiao, Q., Chen, D., Huang, Y., Lin, X., Shen, Y., Li, Y.: Detailmaster: Can your text-to-image model handle long prompts? *CoRR* (2025) [2505.16915](#)
- Stability AI: Stable Diffusion 3. <https://stability.ai/news/stable-diffusion-3-research-paper>. Accessed: 2025-12-04 (2024)
- Relic, L., Azevedo, R., Gross, M., Schroers, C.: Lossy image compression with foundation diffusion models. *ECCV*, 303–319 (2024). Springer
- Esser, P., Rombach, R., Ommer, B.: Taming transformers for high-resolution image synthesis. *CVPR*, 12873–12883 (2021)
- Van Den Oord, A., Vinyals, O., et al.: Neural discrete representation learning. *NeurIPS* **30** (2017)
- Ho, J., Jain, A., Abbeel, P.: Denoising diffusion probabilistic models. *NeurIPS* **33**, 6840–6851 (2020)
- Lu, R., Wang, R., Lyu, K., Jiang, X., Huang, G., Wang, M.: Towards understanding text hallucination of diffusion models via local generation bias. *ICLR* (2025)
- Black Forest Labs: FLUX.1-dev. <https://huggingface.co/black-forest-labs/FLUX.1-dev>. Accessed: 2025-12-04 (2024)
- Liu, X., Gong, C., et al.: Flow straight and fast: Learning to generate and transfer data with rectified flow. *ICLR* (2022)
- Lipman, Y., Chen, R.T., Ben-Hamu, H., Nickel, M., Le, M.: Flow matching for generative modeling. *ICLR* (2023)
- Hu, E.J., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., Chen, W., et al.: Lora: Low-rank adaptation of large language models. *ICLR* (2022)
- Zhang, L., Rao, A., Agrawala, M.: Adding conditional control to text-to-image diffusion models. *ICCV*, 3836–3847 (2023)
- Bentall, R.P.: The illusion of reality: a review and integration of psychological research on hallucinations. *Psychological bulletin* **107**(1), 82 (1990)
- Slade, P.D., Bentall, R.P.: Sensory Deception: A Scientific Analysis of Hallucination. Johns Hopkins series in contemporary medicine and public health. Johns Hopkins University Press, London (1988)
- Ji, Z., Lee, N., Frieske, R., Yu, T., Su, D., Xu, Y., Ishii, E., Bang, Y.J., Madotto, A., Fung, P.: Survey of hallucination in natural language generation. *ACM Comput. Surv.* **55**(12), 1–38 (2023)
- Yu, J., Lin, Z., Yang, J., Shen, X., Lu, X., Huang, T.S.: Generative image inpainting with contextual attention. *CVPR*, 5505–5514 (2018)
- Baker, S., Kanade, T.: Hallucinating faces. *IEEE FG*, 83–88 (2000). IEEE
- Liu, C., Shum, H.-Y., Freeman, W.T.: Face hallucination: Theory and practice. *IJCV* **75**(1), 115–134 (2007)
- Wang, N., Tao, D., Gao, X., Li, X., Li, J.: A comprehensive survey to face hallucination. *IJCV* **106**(1), 9–30 (2014)
- Huang, H., He, R., Sun, Z., Tan, T.: Wavelet domain generative adversarial network for multi-scale face hallucination. *IJCV* **127**(6), 763–784 (2019)
- Quan, W., Chen, J., Liu, Y., Yan, D.-M., Wonka, P.: Deep learning-based image and video inpainting: A survey. *IJCV* **132**(7), 2367–2400 (2024)
- Hariharan, B., Girshick, R.: Low-shot visual recognition by shrinking and hallucinating features. *ICCV*, 3018–3027 (2017)
- Yang, M., Wang, Z.: Image synthesis under limited data: A survey and taxonomy. *IJCV* **133**(6), 3689–3726 (2025)
- Kayhan, O.S., Vredebregt, B., Van Gemert, J.C.: Hallucination in object detection—a study in visual part verification. *ICIP*, 2234–2238 (2021). IEEE
- Piasco, N., Sidibé, D., Gouet-Brunet, V., Demonceaux, C.: Improving image description with auxiliary modality for visual localization in challenging conditions. *IJCV* **129**(1), 185–202 (2021)
- You, J., Shi, H., Jiang, Z., Huang, Z., Gan, R., Wu, K., Cheng, X., Li, X., Ran, B.: V2x-vlm: End-to-end

- v2x cooperative autonomous driving through large vision-language models. *CoRR* (2024) [2408.09251](#)
- Aithal, S.K., Maini, P., Lipton, Z., Kolter, J.Z.: Understanding hallucinations in diffusion models through mode interpolation. *NeurIPS* **37**, 134614–134644 (2024)
- Kim, S., Jin, C., Diethe, T., Figini, M., Tregidgo, H.F., Mullokandov, A., Teare, P., Alexander, D.C.: Tackling structural hallucination in image translation with local diffusion. *ECCV*, 87–103 (2024). Springer
- Wang, H., Cao, J., Liu, J., Zhou, X., Huang, H., He, R.: Hallo3d: Multi-modal hallucination detection and mitigation for consistent 3d content generation. *NeurIPS* **37**, 118883–118906 (2024)
- Saleh, M., Tabatabaei, A.: Building trustworthy multimodal ai: A review of fairness, transparency, and ethics in vision-language tasks. *CoRR* (2025) [2504.13199](#)
- Agnese, J., Herrera, J., Tao, H., Zhu, X.: A survey and taxonomy of adversarial neural networks for text-to-image synthesis. *WIREs DMKD* **10**(4), 1345 (2020)
- Huang, L., Yu, W., Ma, W., Zhong, W., Feng, Z., Wang, H., Chen, Q., Peng, W., Feng, X., Qin, B., *et al.*: A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions. *ACM TOIS* **43**(2), 1–55 (2025)
- Dai, W., Liu, Z., Ji, Z., Su, D., Fung, P.: Plausible may not be faithful: Probing object hallucination in vision-language pre-training. *EACL*, 2136–2148 (2023)
- Nesteruk, S., Svetlana, I., Andrey, S.: Image dataset augmentation a survey and taxonomy. *Measurements and Instrumentation for Machine Vision*, 110–136 (2024)
- Saharia, C., Chan, W., Saxena, S., Li, L., Whang, J., Denton, E.L., Ghasemipour, K., Gontijo Lopes, R., Karagol Ayan, B., Salimans, T., *et al.*: Photo-realistic text-to-image diffusion models with deep language understanding. *NeurIPS* **35**, 36479–36494 (2022)
- Meng, F., Shao, W., Luo, L., Wang, Y., Chen, Y., Lu, Q., Yang, Y., Yang, T., Zhang, K., Qiao, Y., *et al.*: Phybench: A physical commonsense benchmark for evaluating text-to-image models. *CoRR* (2024) [2406.11802](#)
- Hartsock, I., Rasool, G.: Vision-language models for medical report generation and visual question answering: A review. *Frontiers in Artificial Intelligence* **7**, 1430984 (2024)
- Ye, J., Tang, H.: Multimodal large language models for medicine: A comprehensive survey. *CoRR* (2025) [2504.21051](#)
- Zhang, T., Kishore, V., Wu, F., Weinberger, K.Q., Artzi, Y.: Bertscore: Evaluating text generation with bert. *ICLR* (2020)
- Uzunova, H., Ehrhardt, J., Jacob, F., Frydrychowicz, A., Handels, H.: Multi-scale gans for memory-efficient generation of high resolution medical images. *MICCAI*, 112–120 (2019)
- Pinaya, W.H., Tudosi, P.-D., Dafflon, J., Da Costa, P.F., Fernandez, V., Nachev, P., Ourselin, S., Cardoso, M.J.: Brain imaging generation with latent diffusion models. *MICCAI Workshops*, 117–126 (2022). Springer
- Polamreddy, L.R., Roy, K., Yueh, S.-H., Mahato, D., Kuppli, S., Li, J., Zhang, Y.: Leapfrog latent consistency model (llcm) for medical images generation. *CoRR* (2024) [2411.15084](#)
- Heusel, M., Ramsauer, H., Unterthiner, T., Nessler, B., Hochreiter, S.: Gans trained by a two time-scale update rule converge to a local nash equilibrium. *NeurIPS* **30** (2017)
- Hessel, J., Holtzman, A., Forbes, M., Le Bras, R., Choi, Y.: Clipscore: A reference-free evaluation metric for image captioning. *EMNLP*, 7514–7528 (2021)
- Jayasumana, S., Ramalingam, S., Veit, A., Glasner, D., Chakrabarti, A., Kumar, S.: Rethinking fid: Towards a better evaluation metric for image generation. *CVPR*, 9307–9315 (2024)
- Lin, T.-Y., Maire, M., Belongie, S., Hays, J., Perona, P., Ramanan, D., Dollár, P., Zitnick, C.L.: Microsoft coco: Common objects in context. *ECCV*, 740–755 (2014)
- Zhou, B., Zhao, H., Puig, X., Fidler, S., Barriuso, A., Torralba, A.: Scene parsing through ade20k dataset. *CVPR*, 633–641 (2017)
- Krishna, R., Zhu, Y., Groth, O., Johnson, J., Hata, K., Kravitz, J., Chen, S., Kalantidis, Y., Li, L.-J., Shamma, D.A., Bernstein, M.S., Fei-Fei, L.: Visual genome: Connecting language and vision using crowdsourced dense image annotations. *IJCV* **123**, 32–73 (2016)
- Shao, S., Li, Z., Zhang, T., Peng, C., Yu, G., Zhang, X., Li, J., Sun, J.: Objects365: A large-scale, high-quality dataset for object detection. *ICCV*, 8430–8439 (2019)
- Hudson, D.A., Manning, C.D.: Gqa: A new dataset for real-world visual reasoning and compositional question answering. *CVPR*, 6700–6709 (2019)
- Kafle, K., Kanan, C.: An analysis of visual question answering algorithms. *ICCV*, 1965–1973 (2017)
- Kuznetsova, A., Rom, H., Alldrin, N., Uijlings, J., Krasin, I., Pont-Tuset, J., Kamali, S., Popov, S., Mallocci, M., Kolesnikov, A., *et al.*: The open images dataset v4: Unified image classification, object detection, and visual relationship detection at scale. *IJCV* **128**(7), 1956–1981 (2020)

- Wang, X., Peng, Y., Lu, L., Lu, Z., Bagheri, M., Summers, R.M.: Chestx-ray8: Hospital-scale chest x-ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases. *CVPR*, 2097–2106 (2017)
- Liu, B., Zhan, L.-M., Xu, L., Ma, L., Yang, Y., Wu, X.-M.: Slake: A semantically-labeled knowledge-enhanced dataset for medical visual question answering. *ISBI*, 1650–1654 (2021). IEEE
- Bai, Z., Wang, P., Xiao, T., He, T., Han, Z., Zhang, Z., Shou, M.Z.: Hallucination of multimodal large language models: A survey. *CoRR* (2024) [2404.18930](#)
- Agrawal, H., Desai, K., Wang, Y., Chen, X., Jain, R., Johnson, M., Batra, D., Parikh, D., Lee, S., Anderson, P.: Nocaps: Novel object captioning at scale. *ICCV*, 8948–8957 (2019)
- Udandaraao, V., Prabhu, A., Ghosh, A., Sharma, Y., Torr, P., Bibi, A., Albanie, S., Bethge, M.: No” zero-shot” without exponential data: Pretraining concept frequency determines multimodal model performance. *NeurIPS* **37**, 61735–61792 (2024)
- Parashar, S., Lin, Z., Liu, T., Dong, X., Li, Y., Ramanan, D., Caverlee, J., Kong, S.: The neglected tails in vision-language models. *CVPR*, 12988–12997 (2024)
- Yu, Q., Li, J., Wei, L., Pang, L., Ye, W., Qin, B., Tang, S., Tian, Q., Zhuang, Y.: Hallucidoctor: Mitigating hallucinatory toxicity in visual instruction data. *CVPR*, 12944–12953 (2024)
- Leng, S., Zhang, H., Chen, G., Li, X., Lu, S., Miao, C., Bing, L.: Mitigating object hallucinations in large vision-language models through visual contrastive decoding. *CVPR*, 13872–13882 (2024)
- Tong, S., Liu, Z., Zhai, Y., Ma, Y., LeCun, Y., Xie, S.: Eyes wide shut? exploring the visual shortcomings of multimodal llms. *CVPR*, 9568–9578 (2024)
- Wei, H., Kong, L., Chen, J., Zhao, L., Ge, Z., Yang, J., Sun, J., Han, C., Zhang, X.: Vary: Scaling up the vision vocabulary for large vision-language model. *ECCV*, 408–424 (2024)
- Li, Y., Zhou, K., Zhao, W.X., Fang, L., Wen, J.-R.: Analyzing and mitigating object hallucination: A training bias perspective. *CoRR* (2025) [2508.04567](#)
- Xie, C., Liu, T., Jiang, L., Zeng, Y., Shen, Y., Huang, W., Li, J., Xu, X., et al.: Tarac: Mitigating hallucination in lvlms via temporal attention real-time accumulative connection. *CoRR* (2025) [2504.04099](#)
- Zhou, Y., Cui, C., Yoon, J., Zhang, L., Deng, Z., Finn, C., Bansal, M., Yao, H.: Analyzing and mitigating object hallucination in large vision-language models. *ICLR* (2024)
- Wu, J., Shi, Z., Wang, S., Huang, J., Yin, D., Yan, L., Cao, M., Zhang, M.: Mitigating hallucinations in large vision-language models via entity-centric multimodal preference optimization. *ACL* (2025)
- Liu, Q., Shang, C., Liu, L., Pappas, N., Ma, J., John, N.A., Doss, S., Marquez, L., Ballesteros, M., Benajiba, Y.: Unraveling and mitigating safety alignment degradation of vision-language models. *ACL*, 3631–3643 (2025)
- Alonso, I., Azkune, G., Salaberria, A., Barnes, J., Lacalle, O.L.: Vision-language models struggle to align entities across modalities. *CoRR* (2025) [2503.03854](#)
- Zheng, H., Zhang, Z.: Modality bias in lvlms: Analyzing and mitigating object hallucination via attention lens. *CoRR* (2025) [2508.02419](#)
- Xie, Y., Li, G., Xu, X., Kan, M.-Y.: V-dpo: Mitigating hallucination in large vision language models via vision-guided direct preference optimization. *EMNLP*, 13258–13273 (2024)
- Li, S., Qu, J., Zhou, Y., Qin, Y., Yang, T., Zhao, Y.: Treble counterfactual vlms: A causal approach to hallucination. *CoRR* (2025) [2503.06169](#)
- Abaid, A., Farooq, M.A., Hynes, N., Corcoran, P., Ullah, I.: Synthesizing cta image data for type-b aortic dissection using stable diffusion models. *IEEE EMBC*, 1–5 (2024)
- Bui, N.-T., Hoang, D.-H., Trinh, Q.-H., Tran, M.-T., Nguyen, T., Gauch, S.: Nein: Telling what you don’t want. *CVPR*, 3107–3115 (2025)
- Brooks, T., Holynski, A., Efros, A.A.: Instructpix2pix: Learning to follow image editing instructions. *CVPR*, 18392–18402 (2023)
- Sharma, P., Ding, N., Goodman, S., Soricut, R.: Conceptual captions: A cleaned, hypernymed, image alt-text dataset for automatic image captioning. *ACL*, 2556–2565 (2018)
- Conwell, C., Tawiah-Quashie, R., Ullman, T.: Relations, negations, and numbers: Looking for logic in generative text-to-image models. *CoRR* (2024) [2411.17066](#)
- Vice, J., Akhtar, N., Hartley, R., Mian, A.: Exploring bias in over 100 text-to-image generative models. *CoRR* (2025) [2503.08012](#)
- Huang, Z., Zhou, P., Yan, S., Lin, L.: Scalelong: Towards more stable training of diffusion model via scaling network long skip connection. *NeurIPS* **36**, 70376–70401 (2023)
- Chefer, H., Alaluf, Y., Vinker, Y., Wolf, L., Cohen-Or, D.: Attend-and-excite: Attention-based semantic guidance for text-to-image diffusion models. *ACM TOG* **42**(4), 1–10 (2023)
- Phung, Q., Ge, S., Huang, J.-B.: Grounded text-to-image synthesis with attention refocusing. *CVPR*, 7932–7942 (2024)
- Jiang, Y., Gupta, A., Zhang, Z., Wang, G., Dou, Y., Chen, Y., Fei-Fei, L., Anandkumar, A., Zhu, Y., Fan, L.: Vima: General robot manipulation with multimodal prompts. *NeurIPS Workshops* (2022)

- Majumdar, A., Aggarwal, G., Devnani, B., Hoffman, J., Batra, D.: Zson: Zero-shot object-goal navigation using multimodal goal embeddings. *NeurIPS* **35**, 32340–32352 (2022)
- Guo, Z., Yagudin, Z., Lykov, A., Konenkov, M., Tsetserukou, D.: Vlm-auto: Vlm-based autonomous driving assistant with human-like behavior and understanding for complex road scenes. *FLLM*, 501–507 (2024). IEEE
- Xu, M., Huang, P., Yu, W., Liu, S., Zhang, X., Niu, Y., Zhang, T., Xia, F., Tan, J., Zhao, D.: Creative robot tool use with large language models. *CoRR* (2023) [2310.13065](#)
- Shah, R., Martín-Martín, R., Zhu, Y.: Mutex: Learning unified policies from multimodal task specifications. *CoRL* (2023)
- Yuan, Z., Zhang, T., Deng, Y., Zhang, J., Zhu, Y., Jia, Z., Zhou, J., Zhang, J.: Walkvlm: Aid visually impaired people walking by vision language model. *CoRR* (2024) [2412.20903](#)
- Xiao, J., Huang, N., Qiu, H., Tao, Z., Yang, X., Hong, R., Wang, M., Yao, A.: Egoblind: Towards egocentric visual assistance for the blind people. *CoRR* (2025) [2503.08221](#)
- Qian, T., Chen, J., Zhuo, L., Jiao, Y., Jiang, Y.-G.: Nuscenes-qa: A multi-modal visual question answering benchmark for autonomous driving scenario. *AAAI* **38**(5), 4542–4550 (2024)
- Yin, T., Zhou, X., Krahenbuhl, P.: Center-based 3d object detection and tracking. *CVPR*, 11784–11793 (2021)
- Jiao, Y., Jie, Z., Chen, S., Chen, J., Ma, L., Jiang, Y.-G.: Msmdfusion: Fusing lidar and camera at multiple scales with multi-depth seeds for 3d object detection. *CVPR*, 21643–21652 (2023)
- Long, K., Guo, J., Zhang, T., Yu, H., Li, X.: A low-rank method for vision language model hallucination mitigation in autonomous driving. *CoRR* (2025) [2511.06496](#)
- Shin, S., Jeon, S., Kim, J., Kang, G.-C., Zhang, B.-T.: Socratic planner: Inquiry-based zero-shot planning for embodied instruction following. *CoRR* (2024)
- Wang, Z., Liu, A., Lin, H., Li, J., Ma, X., Liang, Y.: Rat: Retrieval augmented thoughts elicit context-aware reasoning in long-horizon generation. *CoRR* (2024) [2403.05313](#)
- Moor, M., Huang, Q., Wu, S., Yasunaga, M., Dalmia, Y., Leskovec, J., Zakka, C., Reis, E.P., Rajpurkar, P.: Med-flamingo: a multimodal medical few-shot learner. *ML4H*, 353–367 (2023)
- Wu, C., Zhang, X., Zhang, Y., Hui, H., Wang, Y., Xie, W.: Towards generalist foundation model for radiology by leveraging web-scale 2d&3d medical data. *Nature Communications* **16**(1), 7866 (2025)
- Chen, J., Gui, C., Ouyang, R., Gao, A., Chen, S., Chen, G.H., Wang, X., Cai, Z., Ji, K., Wan, X., et al.: Towards injecting medical visual knowledge into multimodal llms at scale. *EMNLP*, 7346–7370 (2024)
- Li, C., Wong, C., Zhang, S., Usuyama, N., Liu, H., Yang, J., Naumann, T., Poon, H., Gao, J.: Llava-med: Training a large language-and-vision assistant for biomedicine in one day. *NeurIPS* **36**, 28541–28564 (2023)
- Yan, Q., Yuan, Y., Hu, X., Wang, Y., Xu, J., Li, J., Fu, C.-W., Heng, P.-A.: Medhalltune: An instruction-tuning benchmark for mitigating medical hallucination in vision-language models. *CoRR* (2025) [2502.20780](#)
- Lau, J.J., Gayen, S., Ben Abacha, A., Demner-Fushman, D.: A dataset of clinically generated visual questions and answers about radiology images. *Scientific data* **5**(1), 1–10 (2018)
- Zhao, R., Shi, Z., Zou, Z.: High-resolution remote sensing image captioning based on structured attention. *IEEE TGRS* **60**, 1–14 (2021)
- Liu, F., Chen, D., Guan, Z., Zhou, X., Zhu, J., Ye, Q., Fu, L., Zhou, J.: Remoteclip: A vision language foundation model for remote sensing. *IEEE TGRS* **62**, 1–16 (2024)
- Wang, J., Zheng, Z., Chen, Z., Ma, A., Zhong, Y.: Earthvqa: Towards queryable earth via relational reasoning-based remote sensing visual question answering. *AAAI* **38**(6), 5481–5489 (2024)
- Hu, Y., Yuan, J., Wen, C., Lu, X., Liu, Y., Li, X.: Rsgpt: A remote sensing vision language model and benchmark. *ISPRS JPRS* **224**, 272–286 (2025)
- Wang, D., Zhang, J., Du, B., Xu, M., Liu, L., Tao, D., Zhang, L.: Samrs: Scaling-up remote sensing segmentation dataset with segment anything model. *NeurIPS* **36**, 8815–8827 (2023)
- Du, D., Zhu, P., Wen, L., Bian, X., Lin, H., Hu, Q., Peng, T., Zheng, J., Wang, X., Zhang, Y., et al.: Visdrone-det2019: The vision meets drone object detection in image challenge results. *ICCV Workshops* (2019)
- Li, H., Zhang, X., Qu, H.: Ddfav: Remote sensing large vision language models dataset and evaluation benchmark. *Remote Sensing* **17**(4), 719 (2025)
- Truong, T.-D., Nguyen, H.-Q., Nguyen, X.-B., Dowl- ing, A., Li, X., Luu, K.: Insect-foundation: A foundation model and large multimodal dataset for vision-language insect understanding. *IJCV* **133**(10), 7128–7153 (2025)
- Li, L., Li, J., Chen, D., Pu, L., Yao, H., Huang, Y.: Vllf: A vision-language model based lightweight federated learning framework for smart agriculture. *CoRR* (2025) [2504.13365](#)
- Hong, D., Zhang, B., Li, X., Li, Y., Li, C., Yao, J.,

- Yokoya, N., Li, H., Ghamisi, P., Jia, X., *et al.*: Spectralgpt: Spectral remote sensing foundation model. *IEEE TPAMI* **46**(08), 5227–5244 (2024)
- Zhao, X., Lyu, X., Li, X.: Agrivln: Vision-and-language navigation for agricultural robots. *CoRR* (2025) [2508.07406](#)
- Zhou, G., Hong, Y., Wu, Q.: Navgpt: Explicit reasoning in vision-and-language navigation with large language models. *AAAI* **38**(7), 7641–7649 (2024)
- Chawla, S., Singh, N., Drori, I.: Quantifying and alleviating distribution shifts in foundation models on review classification. *NeurIPS Workshops* (2021)
- Ojha, J., Presacan, O., G. Lind, P., Monteiro, E., Yazidi, A.: Navigating uncertainty: A user-perspective survey of trustworthiness of ai in healthcare. *ACM Transactions on Computing for Healthcare* **6**(3), 1–32 (2025)
- Noorani, S., Kiyani, S., Pappas, G., Hassani, H.: Human-ai collaborative uncertainty quantification. *CoRR* (2025) [2510.23476](#)
- Chakraborty, N., Ornik, M., Driggs-Campbell, K.: Hallucination detection in foundation models for decision-making: A flexible definition and review of the state of the art. *ACM Comput. Surv.* (2025)
- Yan, B., Zhang, J., Yuan, Z., Shan, S., Chen, X.: Evaluating the quality of hallucination benchmarks for large vision-language models. *CoRR* (2024) [2406.17115](#)
- Li, Q., Geng, J., Lyu, C., Zhu, D., Panov, M., Kar-ray, F.: Reference-free hallucination detection for large vision-language models. *EMNLP*, 4542–4551 (2024)
- Song, Y., Qiu, L., Zhang, X., Tang, Z.: Hallucination detection via internal states and structured reasoning consistency in large language models (2025) [2510.11529](#)
- Cafagna, M., Rojas-Barahona, L.M., Deemter, K., Gatt, A.: Interpreting vision and language generative models with semantic visual priors. *Frontiers in Artificial Intelligence* **6**, 1220476 (2023)
- Borji, A.: Qualitative failures of image generation models and their application in detecting deepfakes. *Image and Vision Computing* **137**, 104771 (2023)
- Hertz, A., Mokady, R., Tenenbaum, J., Aberman, K., Pritch, Y., Cohen-Or, D.: Prompt-to-prompt image editing with cross-attention control. *ICLR* (2023)
- Mu, Y., Zhang, Q., Hu, M., Wang, W., Ding, M., Jin, J., Wang, B., Dai, J., Qiao, Y., Luo, P.: Embodiedgpt: Vision-language pre-training via embodied chain of thought. *NeurIPS* **36**, 25081–25094 (2023)
- Shao, H., Qian, S., Xiao, H., Song, G., Zong, Z., Wang, L., Liu, Y., Li, H.: Visual cot: Advancing multi-modal language models with a comprehensive dataset and benchmark for chain-of-thought reasoning. *NeurIPS* (2024)
- Zhao, Q., Lu, Y., Kim, M.J., Fu, Z., Zhang, Z., Wu, Y., Li, Z., Ma, Q., Han, S., Finn, C., *et al.*: Cot-vla: Visual chain-of-thought reasoning for vision-language-action models. *CVPR*, 1702–1713 (2025)
- Rawte, V., Mishra, A., Sheth, A., Das, A.: Defining and quantifying visual hallucinations in vision-language models. *TrustNLP Workshop*, 501–510 (2025)
- Zhou, Y., Fan, Z., Cheng, D., Yang, S., Chen, Z., Cui, C., Wang, X., Li, Y., Zhang, L., Yao, H.: Calibrated self-rewarding vision language models. *NeurIPS* **37**, 51503–51531 (2024)
- Kang, G., Gao, W., Zhan, J.: Evaluatology-driven artificial intelligence. *TBench*, 100245 (2025)
- Rudin, C.: Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. *Nature machine intelligence* **1**(5), 206–215 (2019)
- Yang, Y., Panagopoulou, A., Zhou, S., Jin, D., Callison-Burch, C., Yatskar, M.: Language in a bottle: Language model guided concept bottlenecks for interpretable image classification. *CVPR*, 19187–19197 (2023)
- Ben Melech Stan, G., Aflalo, E., Rohekar, R.Y., Bhiwandiwalla, A., Tseng, S.-Y., Olson, M.L., Gurwicz, Y., Wu, C., Duan, N., Lal, V.: Lvlm-intrepret: An interpretability tool for large vision-language models. *CVPR*, 8182–8187 (2024)
- Wu, K., Xu, S., Chen, H., Wang, C., Li, Z., Wang, Y., Zhong, F.: Vlm can be a good assistant: Enhancing embodied visual tracking with self-improving visual-language models. *CoRR* (2025) [2505.20718](#)
- Sarch, G., Jang, L., Tarr, M., Cohen, W.W., Marino, K., Fragkiadaki, K.: Vlm agents generate their own memories: Distilling experience into embodied programs of thought. *NeurIPS* **37**, 75942–75985 (2024)
- Kalai, A.T., Nachum, O., Vempala, S.S., Zhang, E.: Why language models hallucinate. *CoRR* (2025) [2509.04664](#)
- Yu, X., Cheng, H., Liu, X., Roth, D., Gao, J.: Reeval: Automatic hallucination evaluation for retrieval-augmented large language models via transferable adversarial attacks. *NAACL*, 1333–1351 (2024)
- Zhu, Z., Yang, Y., Sun, Z.: Halueval-wild: Evaluating hallucinations of language models in the wild. *CoRR* (2024) [2403.04307](#)